

**Bilinguals track morphemic co-occurrences better
than quadrilinguals: A novel implicit bilingual
statistical learning paradigm exploring continuous
multilingualism**

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Abstract

While previous bilingual statistical learning experiments found varied bilingual advantages, these were always achieved in the context of an explicit language cue for participants to track regularities separately in both languages. The present work constructs a novel bilingual statistical learning task in which this cue is implicit, and must be implied from monitoring the co-occurrence of morphemes. Over the course of three studies, the paradigm is refined and simplified, and finds evidence for group-level segmentation and co-occurrence tracking to differentiate novel between-language from within-language morphemic pairings. Participants were from a very wide range of language backgrounds, and data for up to four of their spoken languages was used to form multiple continuous predictors. Analysis of individual differences was aided by the development of a more exact language balance measure, which was responsible for the finding of possible quadrilingual difficulties in co-occurrence tracking.

Introduction

Statistical learning

When presented with absolutely novel input, our mind attempts to make sense of it as quickly as possible. This can occur without explicit control, and begins with the automatic and unconscious detection of the underlying statistics within the input. Slowly, over time, the accumulation of lower-order statistical information can be leveraged to form higher-order inferences and conscious knowledge of what we are perceiving. Within the statistical learning community, debate has been ongoing about the quality of statistical learning: whether it is modality-specific or domain-general, whether it is one process applied to different situations or is several different computations intervening at different stages of processing, whether it is aided by other cognitive abilities.

One of the most elegant accounts of the nature of statistical learning comes from [Thiessen et al. \(2013\)](#). They outline that information in novel input holds various statistical regularities, which may be used in different ways to understand the input. Conditional statistics map the frequency-related relationship between elements and determine to which extent one predicts the other. This is what was measured in [Saffran et al. \(1996\)](#)'s pioneering experiment. The continuous speech stream “bidakupadotigolabubidaku...” is made of three-syllable “words”: “bidaku | padoti | golabu...”. Over the course of the experiment, the transitional probabilities (the probabilities of one element following another one; TPs) are computed: within-word syllables “bi”, “da”, and “ku” always follow each other (TP = 1), while between-word syllable TPs are much less reliable (“ku” and “pa” would only have a TP of 0.33). In this way, conditional statistics tell participants that individual syllables form words, and that there are different words in this input. These kind of statistical analyses determine boundaries in stimuli, and allow for them to be stored as separate representations. This is the nature of the process the authors called extraction.

Other types of statistics exist. Distributional statistics, where elements can be grouped categorically according to a central tendency, is a type of learning called integration. This is what operates when doing phonological classification, for example: while there is variation in the detailed acoustic properties of a phoneme, listeners are able to classify all of these as variations around a same theme. Extraction and integration form a complementary system: input is divided into individual clusters through extraction, and grouped together thanks to integration.

In the world of linguistic input, the goal is that of communication, of understanding the message behind the perceived randomness. The mind is highly motivated to learn informative cues that might aid the formation of lexical items. Many aspects of language echo across different existing languages, and the knowledge of other language structures can provide an approximate blueprint of what to search for in other languages. In linguistics, morphology holds one of the keys to decrypting a word. Should a reader not know the meaning of the rare English word “irenic”, the ability to detect the suffix “-ic” within it allows us to conclude that this word is an adjective. Morphemes hold information on part of speech, structure, and semantics of a word. Affixes, through being much higher in frequency than stems, are more familiar to us, and allow us to understand new words. Affix detection and tracking can therefore allow for easier word learning. This is supported by a statistical learning study testing learning of affixation patterns in an artificial lexicon by [Lelonkiewicz, Ktori, et al. \(2020\)](#). Through purely passive exposure to these artificial affixed words, participants were found to be sensitive to the presence of a morpheme, and even more so if it was placed in its learned correct position within the word.

Bilingual statistical learning

Similar statistical learning paradigms have also been developed in bilingual experimental designs. However, this raises additional complications. In a natural bilingual setting, the content of speech is not the only thing that needs to be tracked, but also the language in which the message is communicated. For a bilingual to converse in one language, this may also entail the inhibition of others, and the usage of regularities only belonging to the active language. When speaking with other bilinguals, the active language might change unexpectedly. This code switching may be more fluent or frequent depending on speaker habits and contextual constraints. This requires a listener to continuously track which language is being spoken and detect information indicating a change of language. These cues then need to lead to quick shift in internal language contexts ([Antovich et al., 2018](#)).

The structural sensitivity hypothesis developed in [Kuo et al. \(2012\)](#) posits that bilinguals may be more efficient at the identification of these language-specific cues and of new regularities in linguistic input. This is a crucial ability, as even monolingual input can be rife with variability. How, then, can one determine whether a change in inflection represents a change in language, or merely an accent? There could be two possible sources of variability: variability within the current structure, or variability that indicates a change of structure ([Bulgarelli and Weiss, 2016](#); [Bulgarelli, Lebkuecher, et al., 2018](#)). When mapping the

pattern of regularities they are familiar with, bilinguals therefore need to form multiple representations for their knowledge of the statistics of each of their languages. These, however, may overlap. It could be that bilinguals deploy different strategies to resolve this conflict. Over the course of learning and with much exposure, a bilingual’s statistical representations may become so fine-tuned as to allow even very small variations to act as a cue of a different language (Antovich et al., 2020).

With the importance of language membership cues in mind, some bilingual variations of Saffran et al. (1996)’s experiment have been utilised to examine bilingual segmentation abilities. In Antovich et al. (2018)’s bilingual auditory statistical learning design, words similar to that used in the Saffran et al. (1996) study belonged to one of two languages. One language was spoken entirely by a male voice, and the other by a female voice, and this acted as the language membership cue. Fourteen-month-old infants were exposed to interleaved segments from language 1 and 2. Analysis revealed that only the bilingual infants were able to segment words in both languages. Bilingual infants may be better equipped to track more complex statistical regularities, and/or might do so more efficiently than monolinguals. In a 2019 literature review by Weiss, Schwob, et al. (2020), research from similar experiments indicates that, for interleaved statistical learning, regularities are typically only picked up when they are paired with a clear indexical cue. The risk of not having a clear clue in sequentially-presented languages is to obtain a primacy effect, in which learners only show segmentation abilities in the first language they were given (Bulgarelli and Weiss, 2016).

Another interesting study by Bartolotti, Marian, et al. (2011) examined how similarity of the language membership cues might complicate bilingual learning. In this experiment, artificial Morse languages were presented to participants, with multiple possible segmentation cues present. When these interfered with each other to a high degree, performance for all participants was around chance. However, when there was low interference between cues, bilinguals outperformed monolinguals. Beyond this, the bilinguals with more bilingual experience were better at the task than those with less experience. More bilingual experience might therefore aid in the process of integrating segmentation cues, or to inhibit one of the cues and learn just thanks to the other. This conclusion is similar to that from Weiss, Schwob, et al. (2020), which proposes that prior language experience modulates expectations and the processing of novel artificial languages. Early bilingual experience is thought to have a role in developing inhibitory control, information encoding, the generalisation of memory. Inhibitory control, in particular, could facilitate language context changes (Antovich et al., 2018), and reduce the cognitive load of processing input in two languages concurrently.

However, simple segmentation in a laboratory is far from the level of complexity that assails bilinguals in their usual environment (Frost et al., 2019). Bilingual grammar learning tasks, while few, provide a more difficult task for bilinguals in the lab. One early such study by Nation et al. (1986) found that multilinguals performing an implicit grammar learning task had fewer misses in their responses, and tended to respond more intuitively. Similar ideas were brought up in a paper by Roncaglia-Denissen et al. (2022) on machine classification of bilingual and multilingual performance in a grammar task: multilinguals seemed to perform with more metalinguistic awareness.

Past research therefore seems to suggest a bilingual, or even multilingual advantage for statistical learning tasks that are themselves designed to have a bilingual component. They seem to be able to track patterns that are more complex, and perform some tasks more efficiently.

Limitations to the current literature

However, some important methodological points can be made regarding the analysis of bilingual data. In a great many experiments, participants are themselves chunked as “monolinguals” or “bilinguals”. These labels are very reductive, when the bilingual experience is much richer than the number of languages spoken fluently. Bassetti et al. (2022) declare that studies with bilingual populations should report age of acquisition, proficiency, use, dominance, knowledge of other languages, linguistic and cultural contexts, type of exposure, language of schooling, and length of residence in countries speaking their languages, for a start. This is especially important since it is increasingly difficult to find “true” monolingual participants these days. Especially in countries where English is not the primary cultural language, a lot of that language is picked up online, in films, or on social media. The minimum would be to record their language experience according to the aforementioned metrics to check that they do fulfil the desired “monolingual” characteristics for an experiment. The reality is that these metrics fall on a continuum of multilingualism. Any researcher would be hard-pressed to conclusively determine at which level of second-language experience someone can be deemed a bilingual, and nor would a researcher want to separate two bilingual profiles that sit close together at such a boundary. We therefore need to be using continuous measures to chart multilingual participants.

Differences also exist in how some multilingual metrics are defined and computed. This is notably the case for the measure of multilingual balance. This is applied in two different ways: by thinking about language dominance (which is the “main” language) and language

balance (the extent to which a bilingual’s languages are of equal importance and ability). These are two sides of the same coin, but are not always discussed or computed as such. Language dominance has, in previous studies, been calculated as the difference between a participant’s global language scores (Leclercq et al., 2014; Onnis et al., 2018), or as the ratio of scores (Flege et al., 2002; Sheng et al., 2014). Language balance has sometimes been looked at as the difference only of *proficiency* in various languages (Bartolotti and Marian, 2017). Further complications arise when considering the study of multilinguals, which requires the comparison of more than two numbers. The only solution to this issue that this author has found is that of Gullifer and Titone (2020), who use the notion of entropy to compare multiple language scores to determine balance.

As is perhaps evident from the above sections, the majority of literature on bilingual statistical learning stops at the bilingual participant. However, a similar argument may be used in the determination of bilingualism. How much knowledge of a third language would cause a bilingual to be instead called a trilingual? Our multilingual world is full to the brim with as many different and unique multilingual profiles as there are multilinguals. A reductive approach not only ignores most of that complexity, but also introduces many confounds. As researchers, we may be missing fascinating effects that live purely in this world of messy multilingualism if we stop looking after two languages (Bassetti et al., 2022). To give research every chance of picking up on these, we need to develop and standardise analytical methods that work across the multilingual spectrum.

In past bilingual statistical learning paradigms, the language membership cue only systematically supported learning (at least for bilinguals) when this was very overt. Some studies also explicitly told participants that there were two languages in the artificial input (citeBulgarelli2016), a context in which success may be easier to find (Gebhart et al., 2009). Very perceptually salient cues may also facilitate learning (Weiss, Gerfen, et al., 2009). However, language membership cues such as speaker voice may not have been interpreted by participants as entirely reliable: the same person may speak several languages when interacting with a fellow bilingual. Statistical learning can take place completely unconsciously in the absence of salient cues. It is therefore conceivable that a paradigm can be designed with some form of bilingual statistical learning happening thanks to an entirely implicit, entirely reliable language membership cue, voiding any top-down interferences and prior knowledge.

The current study

The current study tested the use of a completely implicit cue to language membership, that of morphological co-occurrence patterns that do not overlap across two languages. Repeated exposure to artificial words with these patterns should allow for participants to sort the presented words into two separate languages. Participants speaking any number of languages were accepted, and a variety of continuous measures of multilingualism were examined in conjunction with test results to evaluate both the success of the paradigm, and the usefulness of the individual metrics in capturing multilingual profiles.

Experiment 1

Methods

Participants

Participants were 18-25 years of age, had normal or corrected-to-normal vision, and did not have any learning or reading impairments. The study was run online through Prolific ([Prolific 2025](#)), and there were no restrictions on participants’ spoken languages, other than requiring English fluency. To determine the sample size, we ran a simulation-based power analysis. We accounted for 20 items per condition (see below), a 10% difference in responses between conditions, 10% subject intercept variability, 10% subject slope variability, and 15% overall noise. The effect size was set to 5% above chance (see, for example, [Onnis et al. \(2018\)](#), with accuracy in an implicit statistical learning task at 7% above chance). Running 500 simulations, the model estimated that it would be necessary to recruit 200 participants for this effect to be statistically significant in 95% of simulations.

Stimuli

Two chunks of characters formed each word in the experiment. However, given the potentially very wide linguistic background of participants, random assemblies of Latin characters might be meaningful in one of the participants’ languages (e.g. “akcja” is meaningless in English, but is the Polish word for “action”). This is of particular importance, since evidence shows that a participant’s language background and experience influence the processing of new artificial linguistic stimuli ([Siegelman et al., 2018](#); [Weiss, Schwob, et al., 2020](#)), and new statistical structures are not learned in this situation ([Lelonkiewicz, Ullman, et al., 2022](#)). Should some learning occur, any new regularities placed on known lexical tokens would interact with participants’ existing representations ([Frost et al., 2019](#)), and would be an additional confound. This risk was eliminated through the use of the BACS-1 font created by [Vidal et al. \(2017\)](#) for psycholinguistic studies (see Figure 1).

In order to generate artificial languages that are as faithful as possible to real-world language structures, these character chunks were organised as morphemes; similarly to many language systems, the words presented to participants in the present experiment were the assembly of two such morphemes/character chunks (see Figure 1). To ensure that participants were segmenting words according to the morphemes instead of relying on a stable boundary between stem and suffix, stems were 4 or 5 characters long and suffixes 3

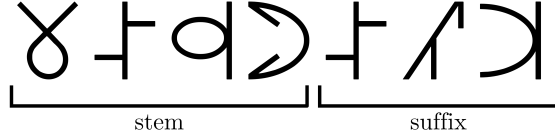


Figure 1: Example of a word in the BACS-1 font (Vidal et al., 2017), composed of a stem and a suffix.

or 4 characters long, thereby forming words 7 to 9 characters long. For each participant, 20 stems and 10 suffixes were randomly generated. We dubbed these chunks “stems” and “suffixes” only based on their distributional properties (e.g. affixes were more frequent and showed up in several different words; see below for details): there was no meaning or syntax in the present experiment, so no stem-affix asymmetry was present in those domains, as would of course happen in real language. Ten stems and five suffixes were then allocated to each of the two stimuli categories, representing two fictional “languages”. As the morphemes used some of the same BACS-1 characters, these languages were visually indistinguishable. This is also an aspect found in natural languages, such as those using the Latin alphabet.

Within each language, all stems were assembled with all suffixes, forming congruent strings (from now on, “words”; see Figure 2). Of these 50 words, 40 were shown as training items, while 10 were kept as testing items. Therefore, throughout the entire set of 80 training words, each stem was present 4 times (stem frequency in training = 0.05) and each suffix was present 8 times (suffix frequency in training = 0.1). The remaining 10 words from each language were taken as congruent testing items. Alongside these, the stems from each language were combined with suffixes from the other language to form 20 incongruent items (see combinations made with dashed lines in Figure 2). These 40 items were therefore named the **two-morpheme** (“2M”) testing condition, with each stem being used twice (stem frequency in 2M = 0.05), and each suffix four times (suffix frequency in 2M = 0.1).

Of note is that, for a participant to identify the two language groups, two different flavours of statistical learning need to be deployed. First, the participant needs to recognise re-occurring morphemes within the words and segment each word into its composing morphemes. Next, co-occurrence tracking across word presentations can identify that these morphemes indeed only co-occur with morphemes from one half of the stimulus set, allowing for the classification of the stimuli into two separate language groups. The learned co-occurrence structure therefore acts as an implicit, fully reliable cue as to stimulus category/language.

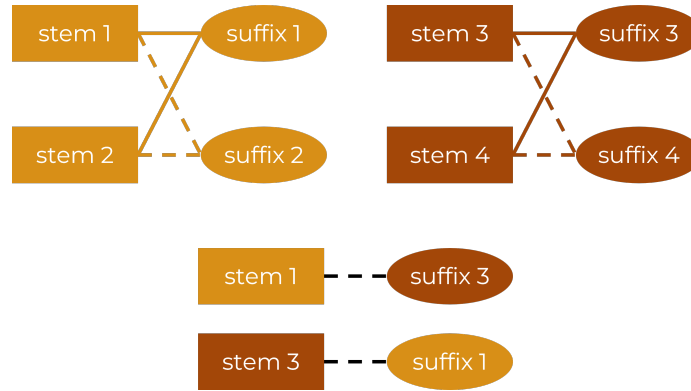


Figure 2: Example of the structure of the training and testing stimuli. In gold: language A, in red: language B. Combinations shown in training are shown by solid connecting lines, and those for testing by dashed connecting lines. This structure is known, for example, by an English-French bilingual. English exposes them to “sing+ing” and “read+ing”, and French gives them “fin+ir” and “chois+ir”, but never “sing+ir” or “chois+ing”: these would seem strange and incompatible with their language experience.

However, the authors had the additional motivation of attempting to replicate and extend the findings of [Lelonkiewicz, Ktori, et al. \(2020\)](#), in which a morpheme familiarity effect was found. In the experiment, participants responded “yes” more often to stimuli containing a familiar morpheme than those containing novel chunks. To achieve this in the current experiment, two additional testing conditions were developed, mirroring those in the original paper. In the first, the **one-morpheme** (“1M”) condition, the morphemes were paired with new random character chunks. These novel chunks were closely matched to the morpheme they replaced. Morpheme position was respected (stems always preceded new chunks, and suffixes were always placed after new chunks), as was chunk length: these had the same number of characters as the morpheme being substituted. Each stem was paired once with a random chunk (stem frequency in 1M = 0.025) and each suffix with 2 different chunks (suffix frequency in 1M = 0.05).

Finally, in the **zero-morpheme** (“0M”) condition, no morphemes were present. Instead, strings made of two novel character chunks were formed.

Overall, there were 40 items in each testing condition. Across all conditions, each stem is shown 3 times (stem frequency = 0.025) and each suffix 6 times (suffix frequency = 0.05).

Familiarity test

In order to evaluate participants’ degree of learning of the training stream, a familiarity test was devised to test whether segmentation of words into morphemes had occurred. If participants were unable to distinguish congruent from incongruent strings in the 2M

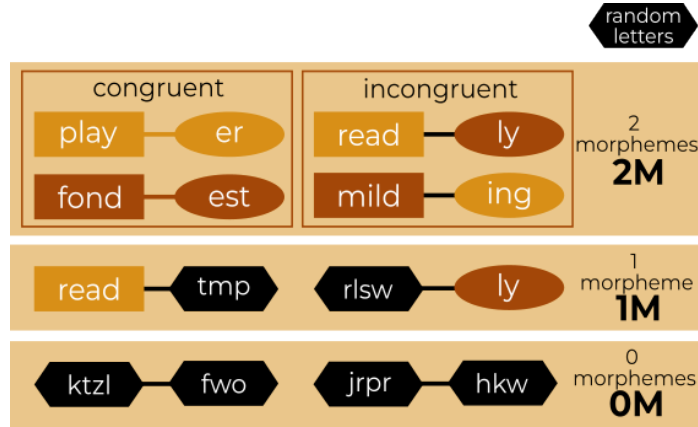


Figure 3: Examples of the structure of testing stimuli in each of the three testing conditions. In gold: language A, in red: language B. In the 2 morphemes (2M) condition, there were novel combinations of same-language (congruent) or different-language (incongruent) morphemes. In the 1 morpheme (1M) condition, the morphemes were combined with novel random strings of letters (black boxes), after [Lelonkiewicz, Ktori, et al. \(2020\)](#). In the 0 morphemes (0M) conditions, novel random letter strings were combined together.

condition, but showed good recognition of the original morphemes in the familiarity test, the failure to perform in the 2M condition could safely be attributed to inadequate co-occurrence tracking instead of poor segmentation (in addition to the comparison between the 1M and 0M conditions).

The items for the familiarity test were, on the one hand, all 30 of the morphemes from both languages present in training and, on the other hand, a set of 30 random chunks distinct from those used in the 0M and 1M conditions. The random chunks were matched for length with the morphemes.

Procedure

Training

Participants were told that they would be acting as language experts needing to pay attention to words received from an alien transmission. Importantly, none of the instructions contained the words “language” or “languages” so as to avoid biasing participants towards either the hypothesis that there was a single language set, or that there were multiple. Instead, phrases such as “the words” or “the alien communication” were carefully chosen to keep this ambiguity: after all, we can’t exclude the possibility that, like humans, aliens might speak multiple languages, even within the same population. Participants were then presented with the training words interleaved between both languages, with each word shown once in each of the 8 randomly shuffled training blocks. Each word was presented for 800ms, with

200ms of inter-stimulus interval. Participants were encouraged to take brief breaks between training blocks to avoid fatigue.

Testing

Participants were then instructed that they would be reviewing words from a new communication from the same group of aliens, and one from a different group of aliens. They were told that both used the same characters, to avoid participants making judgements on this factor. They were told to be quick to respond, pressing “d” or “k” in a 2AFC paradigm to indicate whether or not they believed the string being presented belonged to the first group of aliens. In an interleaved order randomised for each participant, strings from all three (0M, 1M, 2M) conditions were shown. Importantly, in order to probe their implicit knowledge rather than letting them form (necessarily unhelpful) visual classification strategies, a timeout of 2 seconds was set for responding to each item. Should a participant spend more time answering, a screen was shown encouraging them to be quicker. Finally, participants were asked which strategy (if any) they used to respond.

Familiarity test

Following this, the familiarity test was administered. Similarly to the testing phase, participants had the choice of two keys to select which of the two strings presented was the one they had seen previously in the experiment.

Bilingual Language Profile questionnaire

To capture as wide a range of variation within the multilingual population as possible, the Bilingual Language Profile questionnaire (BLP, [Leclercq et al. \(2014\)](#)) was administered. This questionnaire probes language history, use, proficiency, and attitudes, yielding a global language score in a bilingual’s languages. It is listed among the most common tests of language dominance ([Olson, 2023](#)). Test-retest reliability of the questionnaire with a month between sessions yielded excellent reliability of the difference of global scores in two languages (intraclass correlation = 0.979) and also scores of the individual subsections (history ICC = 0.926; use ICC = 0.960; proficiency ICC = 0.881; attitudes ICC = 0.858 - see [Olson \(2023\)](#) for full statistics).

To include participants from all geographical regions and allow for free entry of which languages they speak rather than choosing from the list of possible languages in which the BLP has been translated (which, most notably, completely excludes African languages), it was adapted for participants to respond to these questions for up to four languages they

may speak, whichever these might be. Additionally, an attention check question was added.

Analysis

Pre-processing

Included in the final data sample will be all participants who follow task instructions. Those with a mean response time in testing of over 2 seconds will be excluded, for having replied more on conscious mental strategies over instinctual responses as instructed.

Any data points from testing trials in which the reaction time was inferior to 300ms or superior to 3000ms were excluded from the mixed models analyses.

Participants' BLP subsection and global scores were computed according to the guidelines laid out in [Leclercq et al. \(2014\)](#).

Group-level analyses

In order to evaluate whether the morpheme familiarity effect of [Lelonkiewicz, Ktori, et al. \(2020\)](#) was replicated in this paradigm, a generalized linear mixed-effects model was fitted to evaluate responses between each testing condition, with estimated marginal means calculated for any significant conditions. For performance in the 2M condition, a simple mixed effects model was run to determine group-level performance. An accuracy score for each participant was also computed by summing the hits and correct rejections of each participant, and d-prime scores were extracted. For familiarity test responses, the percentage of correct responses was used in a Student's t-test.

Individual differences analyses

While we strove to deviate from the reductionist tendency to categorise multilinguals according to the number of languages spoken, we also included this variable for the sake of comparison. Based on the 16 scores that that BLP attributes to each participant, we computed the following continuous multilingual metrics.

We used a variable named **multilingual experience** to quantify how much linguistic exposure a participant has across all of their spoken languages. This was measured by summing up all of a participant's language scores.

In continuity with previous research finding language balance to be an important aspect (see [Gullifer and Titone \(2020\)](#)), three measures were elaborated for this construct. **L1-L2 difference** was a measure of **bilingual balance**. This was calculated by taking the absolute

value of the subtraction of the second language (L2) score from the first language (L1) score. Next, **language entropy** was used as a measure of **multilingual balance** that would also consider the effect of eventual third and fourth languages. However, a crucial point in calculating balance is the issue that arises from comparisons between different numbers of languages, so a different number of dimensions of participants' data. Two approaches to resolving this conflict were leveraged. The first, entropy (as in [Gullifer and Titone \(2020\)](#) and [Onnis et al. \(2018\)](#)), was computed on the overall scores for each language via the DescTools R package's entropy function ([Signorelli, 2014](#)). Entropy has the property of applying the same computation to vectors of scores of different lengths. However, a by-product of this is that longer vectors (participants speaking more languages) naturally yield higher entropy scores. As an example, a bilingual speaker with the scores 200 and 200 for their languages has an entropy of 1, while a trilingual speaker with scores of 200, 200, and 200 for their languages receives an entropy score of approximately 1.58, despite both of these profiles showing absolute balance across languages. Entropy might therefore be thought of as a compound measure, considering both multilingual experience and balance.

To untangle multilingual balance from experience, the **cosine similarity** measure was developed (Figure 4).

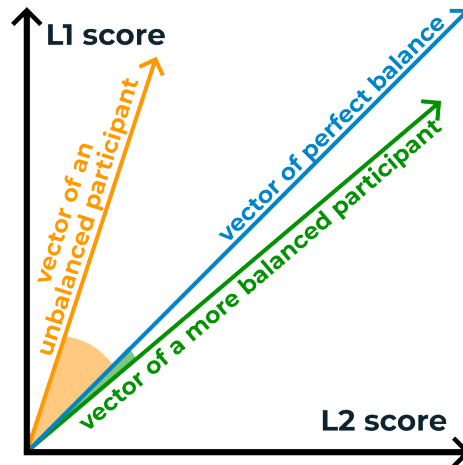


Figure 4: Illustration of the principle of cosine similarity as a measure of language balance. Taking the example of a bilingual participant, one dimension represents the L1 score, with a second dimension for the L2 score. A vector of perfect balance lies in the middle of this two-dimensional plane. Two example participants' language scores are then placed on this plane. First, a participant (in orange) with a high L1 score and low L2 score, is represented as a vector from the origin to the point representing these scores. The angle between this vector and the perfect balance vector is shown in light green. A second example is proposed in green, for a participant with a high L1 score and a slightly higher L2 score. This second participant has a smaller angle to the perfect balance vector, reflecting the closer L1 and L2 scores than in the former example. The same logic can be applied to a higher number of dimensions, which was performed in this study (up to $D = 4$).

Mathematically, with a vector A (the participants' scores vector) and a vector B (the vector of maximal scores):

$$\text{cosine similarity} = \frac{A \cdot B}{\|A\| \|B\|}.$$

A participant vector A is defined as $A = (A_1, A_2, A_3, A_4)$, where A_1 , A_2 , A_3 and A_4 are the overall language scores in L1, L2, L3, and L4, respectively.

Importantly, the perfect vector always has as many dimensions as the participant vector. For example, for a bilingual participant ($A = (210, 180)$ for example), $B = (218, 218)$, whereas for a trilingual participant ($A = (210, 180, 180)$), $B = (218, 218, 218)$.

This metric yields a score between 0 and 1, with 0 for vectors with nothing in common (completely dissimilar, a maximally unbalanced participant), and 1 for identical vectors (a perfectly balanced participant). With the current application, the score range is 0.5 to 1, due to the limitations to how dissimilar the two vectors can be provided that the participant score vector is necessarily composed of two non-null positive values.

However, given that participants were able to declare information on up to four spoken languages no matter their proficiency in that language, one participant may choose, for example, to declare very limited knowledge of Japanese. This could consist in knowing a handful of words. Another participant may choose not to declare such a poor understanding of a language. In the case of the first example participant, a very low score would be introduced into the cosine similarity balance calculation, and pull their score far down. This little knowledge of a language is not likely to affect participants in their daily lives. Therefore, to prevent languages with extremely low scores from affecting overall scores, only scores above or equal to 50 (out of 218) were included in cosine similarity calculations.

Naturally, we couldn't be certain a priori that these metrics would describe our participant sample effectively. Therefore, we also adopted a bottom-up approach and ran a Varimax Principal Components Analysis (PCA) including the BLP scores for each section (L1 History, L1 Use, L2 Proficiency, L2 Use, etc.).

The resulting Principal Components (PCs) were run as predictors in the above linear modelling.

Another aspect of participant background attracted researchers' interest. The script used in each language declared by participants was researched. Participants who were then classified as "same script" users for example spoke English and French, both of which use a

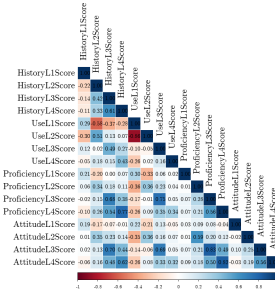


Figure 5: Correlation plot of the BLP variables.

Latin script. Those who spoke for example English and Greek were “different scripts” users.

The metrics were utilised to evaluate individual differences in the morpheme familiarity effect through generalised linear mixed effects models with the following syntax

$$\text{glmer}(\text{observed} \sim \text{scale}(\text{trialn}) + \text{testing condition} * \text{multilingual variable} + (1 + \text{testing condition} | \text{subject ID}), \text{family} = \text{“binomial”})$$

These models were run on all datapoints except those with a response time superior to 300ms and inferior to 3000ms (thereby discarding abnormally short or abnormally long trials).

Mixed effects modelling of 2M performance was achieved through limiting the previous analysis to the 2M condition. In these models, the outcome variable would now be the nature of the strings presented to participants: “within-language” or “between-language”, which are also the options for participant responses:

$$\text{participant response} \sim \text{scale}(\text{trialn}) + \text{string nature} * \text{variable} + (1 + \text{expected} | \text{subject ID})$$

Finally, further mixed effects modelling observed the effect of the variables on participants’ familiarity test score. These models were formulated in the same way as for 2M performance.

All linear modelling was achieved through usage of the lme4 (Bates et al., 2015) package in R.

Results

Participants

After participant exclusions, 193 participants were retained in the final sample. Of these, 112 were male. Participants' mean age was 22.96 years.

Testing error

Unfortunately, due to a technical issue, the fifth round of training showed participants a single word, repeated eighty times. Due to this over-familiarisation, the word as well as its composing morphemes was excluded from all subsequent analyses.

Group-level analyses

Familiarity test

Participants significantly chose known morphemes above matched controls in an estimated 57% of trials ($t(192) = 9.16$, $p < 2.2e^{-16}$).

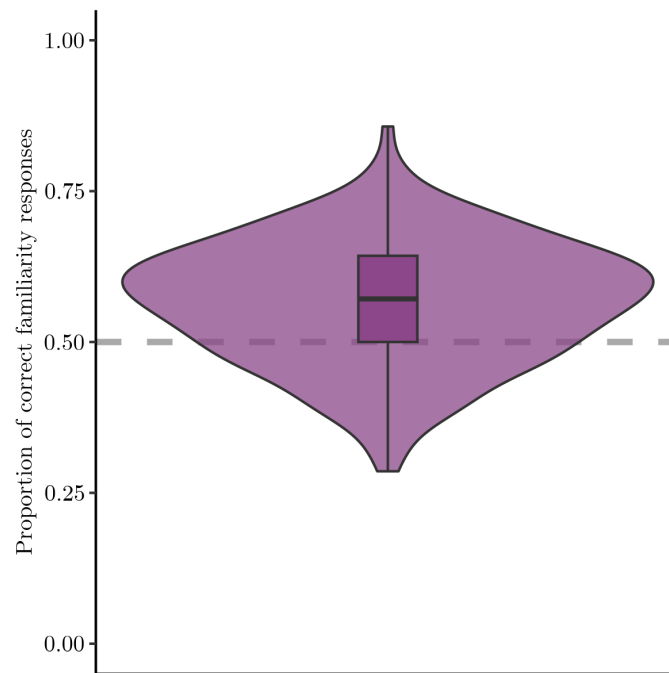


Figure 6: Violin plot and boxplot of the accuracy of familiarity test responses. The dashed line indicates chance level.

Morpheme familiarity effect

Participants' responses across testing conditions is represented in Figure 7. Linear modelling showed significant differences between each testing condition. The proportion of “within-language” responses in the 0M condition didn't significantly differ from chance (Intercept $\beta = -0.05$, $z = -1.06$, $p = 0.29$). There were significantly more “within-language” responses in the 1M condition ($\beta = 0.28$, $z = 7.22$, $p = 5.31e^{-13}$), and even more in the 2M condition ($\beta = 0.67$, $z = 12.74$, $p = 3.402e^{-37}$). Estimated marginal means analysis estimated the proportion of “within-language” responses to be 49% for 0M, 56% for 1M, and 65% for 2M.

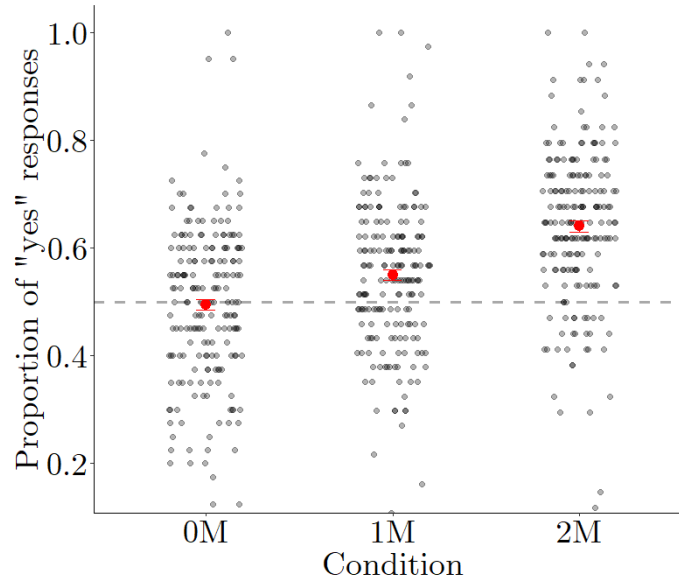


Figure 7: Proportion of “yes” responses in each testing condition. The dashed line indicates chance level. The red circle shows the group mean, with bars as the standard error of the mean.

2M performance

Overall, performance in the 2M condition was not significantly different from chance ($t(192) = -0.64$, $p = 0.53$). D-prime analysis yielded an average c of -0.37 ($t(192) = -12.39$, $p < 2e^{-16}$) and average d -prime of -0.02 ($t(192) = -0.76$, $p = 0.45$): while participant responses were biased towards “yes”, a d -prime so close to zero indicates no detection of the underlying structure. Response times and accuracy in the 2M condition were uncorrelated ($r(191) = 0.04$, $p = 0.58$), meaning there was no speed-accuracy trade-off here.

Individual differences analyses

BLP

Lots of possible vars - after exp1 data, drop vars that are too similar so simplify down

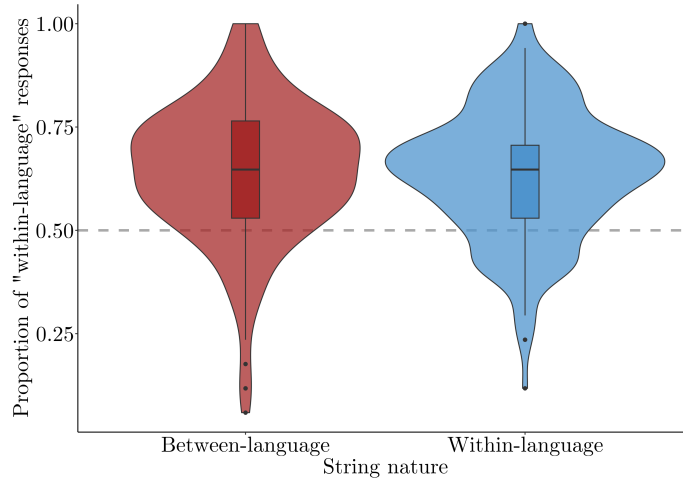


Figure 8: Boxplots of the proportion of “within-language” responses to each 2M combination type. The dashed line indicates chance level.

num of vars looked at

In the BLP, 5 participants reported information on just one spoken language, 83 for two, 56 for three, and 49 for four. Of note is that the BLP instructions permitted reporting of languages spoken with any proficiency at all, so these divisions cannot be thought of as fluent monolingual/bilingual/trilingual/quadrilingual participants. The most popular L1 was Portuguese, the most popular L2 was English, the most popular L3 was Spanish, and the most popular L4 was French. Six unique language families were present in this sample, with a total of ten unique branches, and thirty-nine total languages. Out of these participants, 173 spoke languages with the same script, and 20 with different scripts.

The Pearson correlation between entropy and multilingual experience scores indicated that these metrics are very strongly positively correlated: $t(191) = 37.30$, $p < 2e^{-16}$, estimated $r = 0.94$. On the other hand, the correlation between cosine similarity and multilingual experience shows these measures as distinct: $t(191) = -1.68$, $p = 0.09$, estimated $r = -0.12$. This provides evidence for the necessity of the cosine similarity measure as a multilingual metric separate from the influence of the number of languages contributing to its calculation.

PCA

The PCA extracted several PCs that will be used in subsequent analyses. PC1 was composed of L3 components, and makes up 17% of the variance in scores. PC9 was the L4 equivalent, worth 13% of variance. PC2 was of particular interest to the authors, as it strongly negatively

weighs L1 use against strongly positively weighted L2 use (12% of variance explained). PC8 loaded L2 history (how early participants speak a language well, 6% of variance), while RC3 loaded L2 proficiency (6% of variance). The resulting scree plot can be found in the Supplementary Materials.

Familiarity test

No variables significantly influenced familiarity responses.

Morpheme familiarity effect

There were significant interactions of all levels of multilingual category with the different testing conditions compared with the intercept (monolinguals). However, given the extremely low number of observations of monolinguals ($n = 5$), these effects were deemed unreliable.

PC8 (L2 history) was significant as a main effect on “yes” responses ($\beta = 0.11$, $z = 2.38$, $p = 0.02$), with longer L2 history prompting more “yes” responses. Earlier bilinguals might therefore be predisposed to accept a wider range of stimuli as linguistically-coherent. Older participants accepted fewer combinations overall ($\beta = -0.10$, $z = -2.21$, $p = 0.03$), an effect seemingly driven by an increase in rejections of the 0M novel combinations. Indeed, the morpheme familiarity effect was stronger for older participants (Age*1M interaction: $\beta = 0.07$, $z = 1.92$, $p = 0.054$; Age*2M interaction: $\beta = 0.14$, $z = 2.77$, $p = 0.006$). Adults with more exposure to different language systems over the course of their life may be more reliant on previously seen chunks.

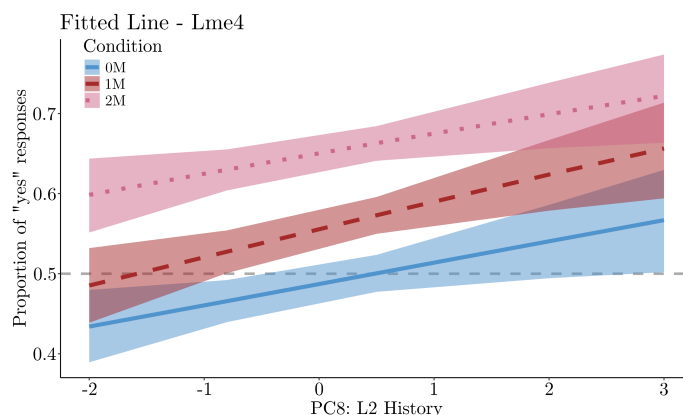


Figure 9: Model estimates of the effect of PC8 (L2 History) on “yes” responses, for the 0M (blue), 1M (red), and 2M conditions (pink). Higher values of PC8 translates to earlier experience and abilities in L2. Bars around the lines indicate the 95% confidence interval. Dashed lines at 0.5 indicate chance level.

PC3 (L2 proficiency) also strengthened the morpheme familiarity effect (PC3*1M

interaction: $\beta = 0.08$, $z = 2.24$, $p = 0.03$; PC3*2M interaction: $\beta = 0.15$, $z = 2.91$, $p = 0.004$). This was also the case for L1-L2 difference, which interacted significantly with the 1M and 2M conditions (L1-L2 difference*1M interaction: $\beta = -0.10$, $z = -2.81$, $p = 0.005$; L1-L2 difference*2M interaction: $\beta = -0.13$, $z = -3.65$, $p = 0.0003$). It may be that those with a more fluent and balanced interaction of their languages are better able to use past knowledge about new stimuli.

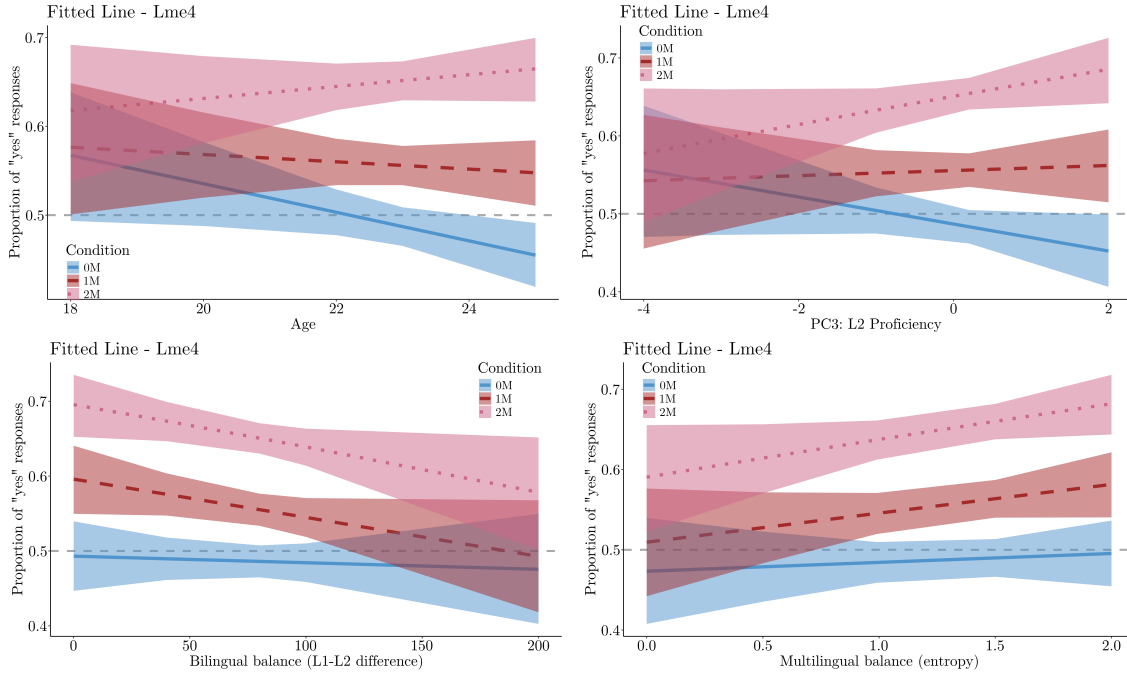


Figure 10: Model estimates of the effect of Age (top left), PC3 (L2 Proficiency; top right), bilingual balance (L1-L2 difference; bottom left), and multilingual balance (entropy; bottom right) on “yes” responses, for the 0M (blue), 1M (red), and 2M conditions (pink). Higher values of PC3 entail higher L2 proficiency. L1-L2 difference scores closer to 0 indicate a more balanced bilingual profile, while entropy values closer to 2 indicate a more balanced multilingual profile. Bars around the lines indicate the 95% confidence interval. Dashed lines at 0.5 indicate chance level.

“Yes” responses across conditions				
	β	SE	z-value	p-value
PC8 (L2 history)	0.11	0.04	2.38	0.02
Age	-0.10	0.04	-2.21	0.03
Age*2M	0.14	0.05	2.77	0.006
PC3 (L2 proficiency)*1M	0.08	0.04	2.24	0.03
PC3 (L2 proficiency)*2M	0.15	0.05	2.91	0.004
L1-L2 difference*1M	-0.10	0.04	-2.81	0.005
L1-L2 difference*2M	-0.13	0.04	-3.65	0.0003

Table 1: Summary of experiment 1 mixed modelling statistics of “yes” responses across conditions, for significant variables.

2M performance

Participant strategies were classified, the most popular of these being the use of one’s intuition (declared by 80 participants). The next most used strategies were single letter identification (56 participants) and morpheme identification (25 participants). Given that intuition would rely on participants’ implicit knowledge, a Student’s t-test was also run on those participants’ 2M scores. However, this revealed that their scores were not significantly above chance ($t(79) = -0.94$, $p = 0.35$).

On the other hand, while those using the learned morphemes as a strategy did not have scores significantly above chance ($t(24) = 1.01$, $p = 0.32$), they did have a mean d-prime of 0.10. Given that no other variables significantly affected 2M accuracy, this was a good indicator that the task may be too complicated, and only those particularly looking for meaningful chunks in the input were able to successfully detect when they were assembled incorrectly.

Discussion

This experiment piloted a new statistical learning paradigm to study morpheme learning in an implicit, bilingual setting. A familiarity test demonstrated that participants did indeed pick up some basic knowledge of the morphemes they were exposed to in the initial training phase. An instinctive bias towards familiarity also showed up in the distribution of “yes” responses across the testing conditions: as was observed in [Lelonkiewicz, Ktori, et al. \(2020\)](#), participants preferred to label as congruent any strings that contained a chunk that they knew already; this effect is amplified when two known chunks are present in the proposed strings. This replication of previous work laid a solid foundation on which to extend that work into the domain of individual variability.

One principal aim of the present research has been a wider inclusion of individual variability than what has been sometimes sought after for the sake of limiting sample noise. The success of our approach was seen in the diversity of backgrounds and languages measured. A PCA done on participants’ language scores did not account for as much variance as might have been hoped for, but, given the astonishing number of differences in language learning and habits, it can be said that this uniqueness was partly distilled into a few useful data-driven components. Additionally, the novel cosine similarity metric was shown to be valuable as a “pure” measure of language balance that is not affected by the number of languages spoken, unlike the usual entropy metric.

While none of these bottom-up, or indeed none of the top-down measures either, seemed to influence the segmentation abilities of participants, the PCA proved an important tool when examining the pattern of “yes” responses to the testing items. The earlier one starts learning a second language, the more prone they are to accept novel stimuli as being linguistic in nature.

However, some factors also influence a more measured sensitivity to new combinations involving word parts they already know. Age was found to induce a larger morpheme familiarity effect, whereby the mere presence of a known linguistic chunk is enough to suggest to a participant that this novel assembly is congruent with their past knowledge. Participants with more sensitivity to this (namely, older participants) respond “yes” at chance level for chunks with no known parts, slightly more to those containing one known part, and even more for those being a combination of two known parts. This age-related enhancement may be due to their longer time manipulating linguistic systems, leading to their having seen a wider variety of combinations of morphemes, and therefore more reliance

on known structures. This can also be said about the significant interaction between L2 proficiency and this morpheme familiarity effect. More proficient language users may have a larger vocabulary, and a similar reason for being more sensitive to the presence of familiar word parts.

On a slightly different plane, one further measure pushed the morpheme familiarity effect for certain participants. Those with a smaller difference in scores between their first and second languages also exhibited this pattern of “yes” responses. These would be participants who make use of their top two languages more equally, and can be expected to be more flexible in working with both. This flexibility could explain the easier acceptance of new words linked to those previously stored in memory, especially if they’re linked to several different part-word items.

However, a solid null result was observed on a group scale of the discrimination performance between within- and between-language morphemes combinations in condition 2M. Crucially, one group show some aptitude for correct sorting of stimuli in the 2M condition. Those having reported using known morphemes in the testing strings seemed better equipped to reject between-language items than those using other strategies or their intuition. These participants were using a morphology-based classification strategy: they may have understood the underlying structure better than participants who focused more on letter-based statistics. A promising avenue for improving the experimental design might be to boost morpheme learning.

Experiment 2

Methods

Participants

All participant criteria were identical to experiment 1.

Stimuli

All stimuli parameters were identical to experiment 1.

Procedure

Morpheme spelling task

Given that the current task requires initial segmentation and subsequent co-occurrence tracking, and that the latter is the process we were most interested in, a new task was devised with the aim of facilitating segmentation. This would allow participants to spend fewer cognitive resources on this step, hopefully allowing for better co-occurrence tracking. Therefore, a familiarisation task with all the morphemes presented in training was given as the first task in the experiment.

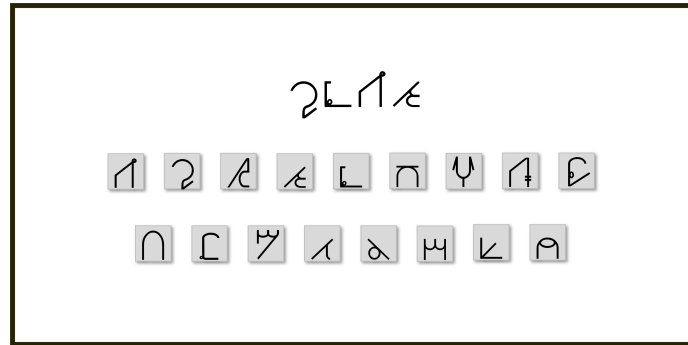


Figure 11: Example item of the morpheme spelling task. A target morpheme is shown above with all of the individual letters used in the experiment in a keyboard below. A participant must click each letter to form the target morpheme.

In order to ensure that participants were actively learning the morphemes rather than being passively exposed to them, a spelling task was created (illustrated in Figure 11). This procedure would also help participants get used to the unfamiliar BACS-1 characters. Participants were presented with each morpheme that was to be used in the experiment, and had to click on each composing letter on the on-screen BACS-1 keyboard. Each morpheme

was displayed until the participant completed its spelling. We introduced each of the 30 morphemes once.

Training

The aforementioned coding error was resolved, presenting participants 8 true training blocks.

Analysis

This time, the analysis included all items in each of the testing conditions, and all morphemes in the familiarity test. All data was analysed in precisely the same way as in experiment 1.

Results

Participants

One hundred and ninety one participants completed the experiment. Seven participants were then excluded for the following reasons: having a very slow mean testing response time (mean = 1986ms, $n = 1$ participant), having a very fast mean testing response time (mean = 175ms, $n = 1$), declaring to have filmed the training phase to help performance in testing ($n = 1$), having never responded “yes” in testing ($n = 1$), and having always responded “yes” in testing ($n = 3$). Therefore, data from 184 participants were analysed. This sample counted 93 males, and had a mean age of 22.22 years.

Group-level analyses

Familiarity test

This experiment also saw mean familiarity test scores significantly above chance ($t(183) = 10.2$, $p < 2.2e^{-16}$) at 57%. While the mean scores were comparable to experiment 1, responses in this phase were 40ms quicker than in experiment 1, indicating a better grasp of the material.

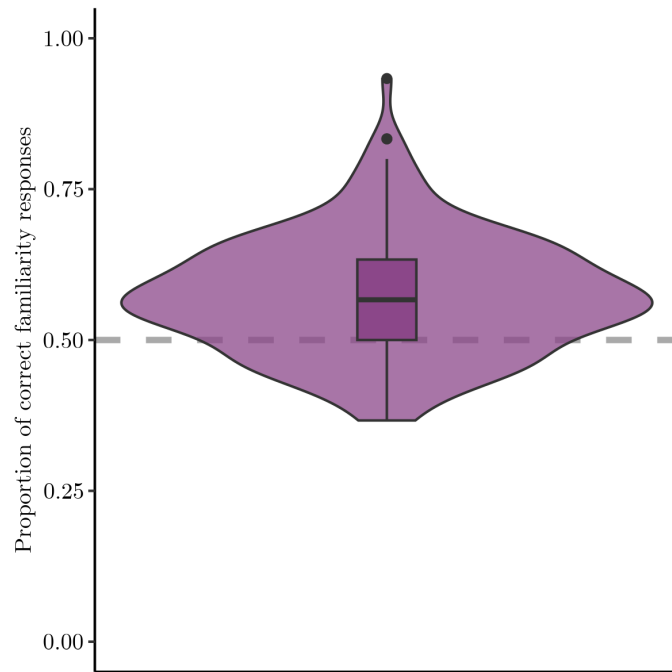


Figure 12: Violin plot and boxplot of the accuracy of familiarity test responses. The dashed line indicates chance level.

Morpheme familiarity test

Results from experiment 1 were successfully replicated, with the morpheme familiarity effect also being present in experiment 2. Significant differences were found between the 0M (model intercept), 1M, and 2M conditions after linear modelling: Intercept $\beta = 0.09$, $z = 1.74$, $p = 0.08$; 1M $\beta = 0.25$, $z = 5.81$, $p = 6.30e^{-9}$; 2M $\beta = 0.55$, $z = 10.77$, $p = 5e^{-27}$. Participants confronted with the 1M condition responded “yes” 6% more than in the 0M condition (0M model fit = 0.52, SE = 0.01; 1M model fit = 0.58, SE = 0.01). The 2M condition attracted 7% more “yes” responses than the 1M condition (2M model fit = 0.65, SE = 0.01). This pattern of results replicates that found in experiment 1.

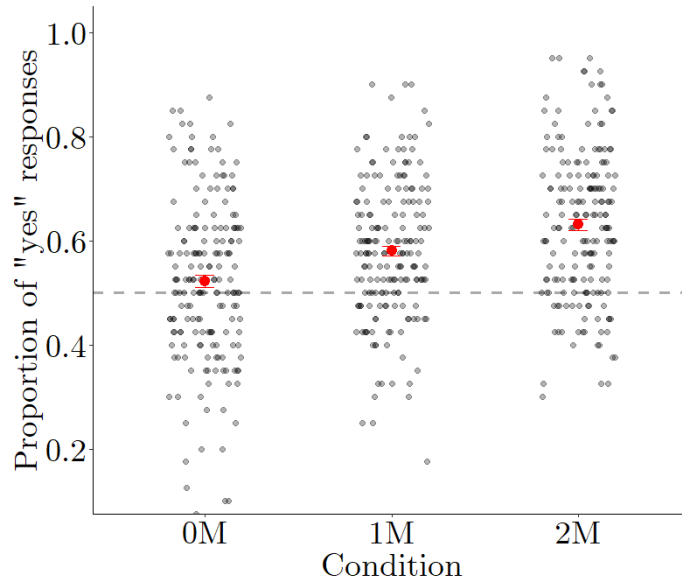


Figure 13: Proportion of “yes” responses in each testing condition. The dashed line indicates chance level. The red circle shows the group mean, with bars as the standard error of the mean.

2M performance

However, mean 2M accuracy was still not significantly different from chance: $t(183) = -0.81$, $p = 0.42$. In this version of the experiment, the mean d' -prime score was -0.03 , with an average c of -0.35 , still showing no co-occurrence learning despite a clear bias towards “yes” in the 2M condition.

Individual differences analyses

BLP

Twenty-two participants completed the BLP for just one language, 70 did so for two

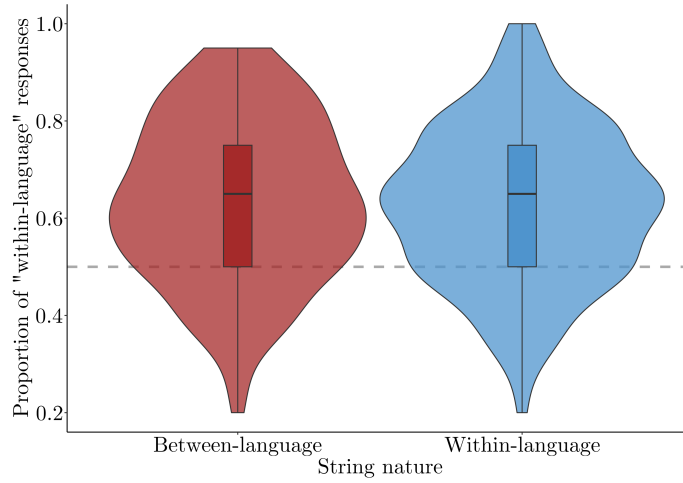


Figure 14: Boxplots of the proportion of “within-language” responses to each 2M combination type. The dashed line indicates chance level.

languages, 51 for three, and 41 for four languages. English was the most dominant language, being the top L1 and L2 reported by participants. German was the most common L3 language, and Spanish was the most common L4. The languages declared by participants belonged to a total of 11 language families, with 20 unique branches. In this version of the experiment, 156 participants were same script multilinguals, and 28 were different script multilinguals.

PCA

The PCA conducted on experiment 2 data produced five interesting PCs. PC1 loaded L4 components, and represented 20% of the variance in the sample (the equivalent of PC9 from experiment 1). PC3 explained 17% of the variance by representing L3 information (like PC1 in experiment 1). PC2 presented the contrast between L1 and L2 language use, with L1 use strongly negatively weighted, and L2 use strongly positively weighted (also like experiment 1’s PC2). PC7 was for the L2 history measure (5% of variance), replacing PC8 in experiment 1 logging the same variable. Finally, in an interesting deviation from experiment 1, no L2 proficiency measure was kept by the PCA in experiment 2, which instead placed value on PC9 storing L4 use (5% of variance).

Familiarity test

The PC component 9 dealing with L4 use was the only measure to significantly affect familiarity test accuracy: $\beta = -0.07$, $z = -2.30$, $p = 0.02$.

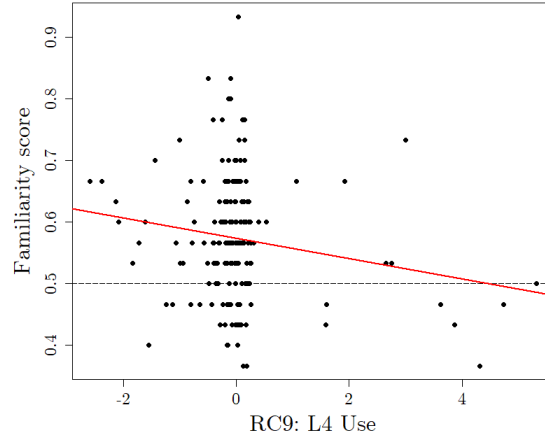


Figure 15: Familiarity test performance according to PC9 for L4 use, with model regression line in red. The dashed line at 0.5 indicates chance level.

Morpheme familiarity effect

Several variables influenced the overall proportion of “yes” responses across all testing conditions. These were the PC7 for L2 history ($\beta = 0.11$, $z = 2.20$, $p = 0.03$), bilingual balance (L1-L2 difference; $\beta = -0.10$, $z = -2.02$, $p = 0.04$), and multilingual experience ($\beta = 0.15$, $z = 3.34$, $p = 0.0008$).

As compared with the 0M condition, multilingual experience also had a significant interaction effect on 1M “yes” responses ($\beta = -0.08$, $z = -2.26$, $p = 0.02$). The amount of “yes” answers to the 1M condition was also influenced by cosine similarity (1M*cosine similarity interaction: $\beta = 1.72$, $z = 2.05$, $p = 0.04$), with increased multilingual balance meaning greater preference for novel 1M combinations when compared to 0M responses.

Other metrics increased the observed morpheme familiarity effect. The distribution of “yes” responses across the 0M, 1M, and 2M conditions was more discriminate for older participants ($\beta = 0.10$, $z = 2.01$, $p = 0.04$), and for those who used their L2 more than their L1 (PC2; $\beta = 0.10$, $z = 2.03$, $p = 0.04$).

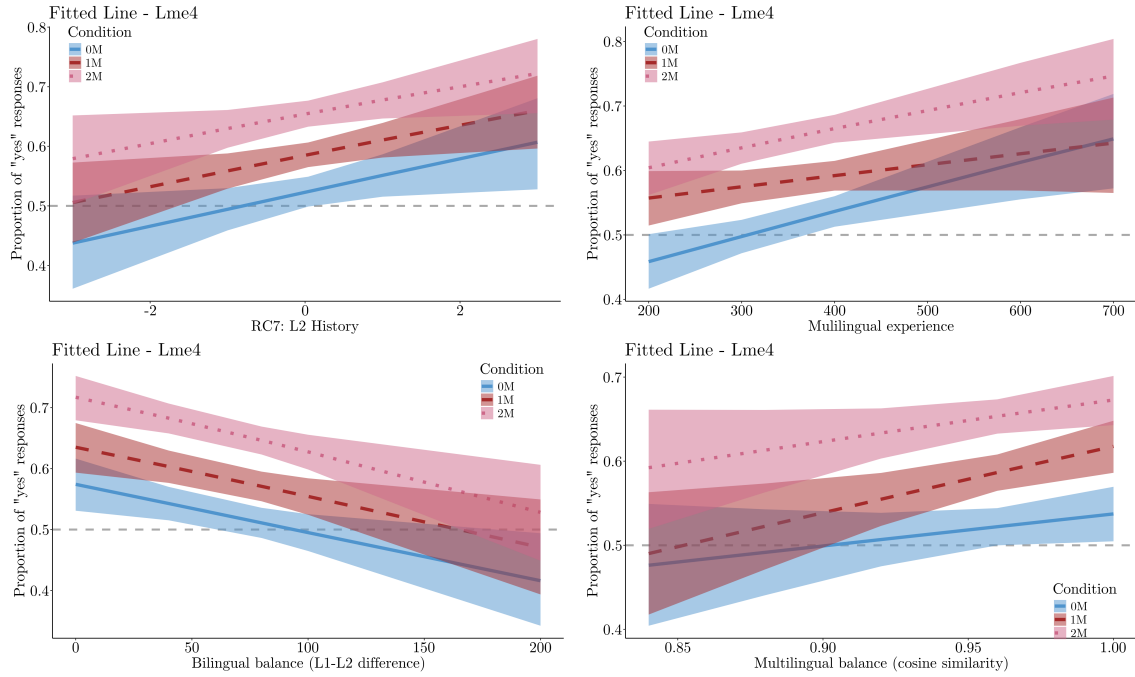


Figure 16: Model estimates of the effect of PC7 (L2 History; top right), multilingual experience (bottom right), bilingual balance (L1-L2 difference; bottom left), and cosine similarity (bottom right) on “yes” responses, for the 0M (blue), 1M (red), and 2M (pink) conditions. Higher values of PC7 means longer L2 knowledge. L1-L2 difference scores closer to 0 indicate a more balanced bilingual profile. Cosine similarity scores closer to 1 indicate more balance across a participants’ languages. Bars around the lines indicate the 95% confidence interval. Dashed lines at 0.5 indicate chance level.

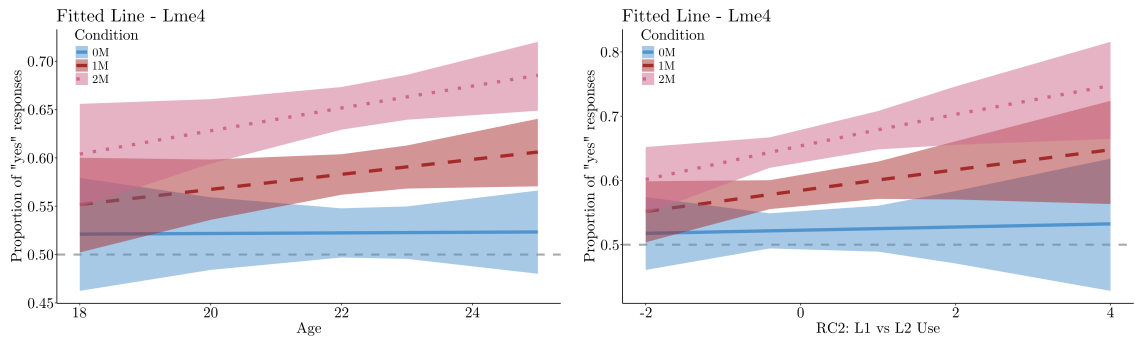


Figure 17: Model estimates of the effect of age (left) and PC2 (L1 vs L2 use; right) on “yes” responses, for the 0M (blue), 1M (red), and 2M conditions (pink). Higher values of PC2 translates to more use of participants’ L2 comparatively to their L1. Bars around the lines indicate the 95% confidence interval. Dashed lines at 0.5 indicate chance level.

“yes” responses across conditions				
	β	SE	z-value	p-value
PC7 (L2 history)	0.11	0.05	2.20	0.03
Bilingual balance (L1-L2 difference)	-0.10	0.05	-2.02	0.04
Multilingual experience	0.15	0.04	3.34	0.0008
Multilingual experience*1M	-0.08	0.04	-2.26	0.02
Age*2M	0.10	0.05	2.01	0.04
PC2 (L1 vs L2 use)*2M	0.10	0.05	2.03	0.04

Table 2: Summary of experiment 2 mixed modelling statistics of “yes” responses across conditions, for significant variables.

2M performance

Measures of balance affected the amount of “yes” responses to the 2M conditions. Higher levels of bilingual imbalance (according to L1-L2 difference) drove the proportion of “yes” responses down ($\beta = -0.15$, $z = -2.69$, $p = 0.007$), and the same was true for multilingual balance (as defined by cosine similarity; $\beta = 0.14$, $z = 2.45$, $p = 0.01$), as can be seen in Figure 18.

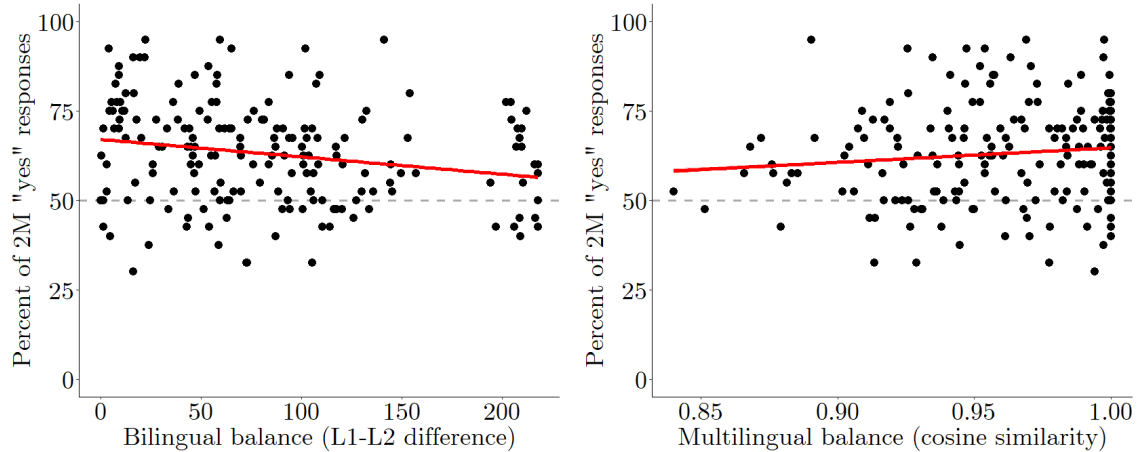


Figure 18: Plots of the percent of “yes” responses in the 2M condition as a function of bilingual balance (L1-L2 difference; left) and multilingual balance (cosine similarity; right). Lower levels of L1-L2 difference indicate more balanced profiles, as do higher levels of cosine similarity. Dashed lines at 50% indicate chance level.

While quadrilinguals had higher 2M accuracy than monolinguals ($\beta = 0.45$, $z = 2.38$, $p = 0.02$), they were not actually above chance ($t(40) = 1.28$, $p = 0.21$; see Figure 19). Multilingual balance, or cosine similarity, also initially had a significant interaction with

2M accuracy ($\beta = -0.11$, $z = -2.13$, $p = 0.03$), but this effect was lessened to the extent of non-significance when the monolingual population was removed from analysis ($\beta = -0.08$, $z = -1.42$, $p = 0.16$). Monolinguals are a special case when it comes to multilingual balance, as they necessarily calculate as perfectly balance due to their single data point. This was therefore deemed insufficient evidence in favour of a multilingual balance effect.

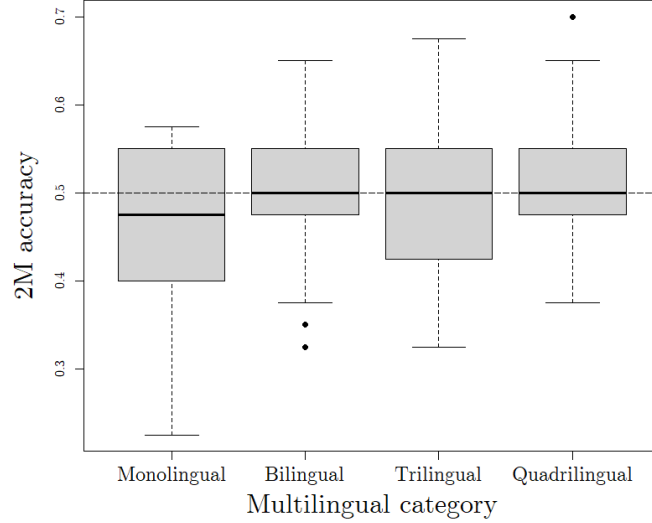


Figure 19: Plot of the relationship between 2M accuracy and multilingual category. The dashed lines indicates chance level.

An interesting relationship between 2M scores and participant language script type was found. Participants who spoke different scripts had 2M scores above chance level ($t(27) = 2.10$, $p = 0.045$), with scores estimated around 53% (Figure 20).

2M “yes” responses and accuracy				
	β	SE	z-value	p-value
Bilingual balance (L1-L2 difference)	-0.15	0.06	-2.69	0.007
Quadrilingual*correct	0.45	0.19	2.38	0.02
Different scripts*correct	0.32	0.15	2.12	0.03

Table 3: Summary of experiment 2 mixed modelling statistics of 2M “yes” responses and accuracy across conditions, for significant variables.

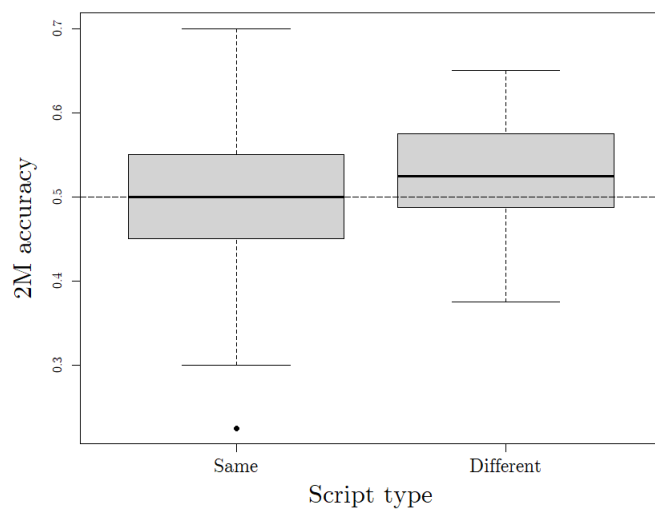


Figure 20: Box plots of 2M accuracy scores for each language script type. The dashed line indicates chance level.

Discussion

Experiment two provided a useful replication of experiment one, with some extra nuance present in participant responses. The addition of a morpheme spelling task prior to any training seemed to have supported morpheme learning a touch. Familiarity with the study stimuli decreased reaction times without compromising accuracy, indicating greater ease with the material. This was affected by the proportion of time participants spent speaking in a fourth language. In the absence of further information, it is unclear to the authors as to why this relationship might exist.

The morpheme familiarity effect was also present in this version of the experiment, and this was modulated by multilingual experience. This measure increased the overall proportion of “yes” responses, indicating that the more experience one has with the wide range of linguistic combinations existing across languages, the more likely a new combination is judged as coherent, even if this combination does not include any previously known chunk. This was also the case with the L2 History PC component, which showed that the earlier one becomes a bilingual, the more all linguistic stimuli seem plausible. This effect was also present in participants more balanced across their first two languages. Frequent and flexible interaction of one’s two main languages seem to increase their acceptance of new linguistic stimuli. Multilingual balance also played a role in participant responses: participants who scores higher on cosine similarity also were better able to differentiate the different testing conditions, and accept significantly more combinations of new and unfamiliar chunks.

Mirroring results from experiment one, experiment two also saw mean 2M responses at chance, indicating that participants on the whole were unable to pick apart congruent and incongruent 2M combinations. However, participants speaking language with multiple different types of scripts did perform above chance. Due to their experience manipulating linguistic items in different alphabets, they may have had an easier time picking up on BACS-1 stimulus characteristics. This may have more similarities to them as a foreign language script rather than something completely alien, which may have been the experience of same-script multilinguals.

This prompted further reflection on the study design. While the original stimulus set had been conceived as similar to true languages, and then converted into BACS-1 out of the necessity to erase any resemblance to any true linguistic chunk, this manipulation could be reserving more cognitive resources for the simple identification of novel letters, and take away from the processing of the morphemic chunks. Only those used to working with foreign

scripts were efficient enough in single letter processing to devote enough resources to the co-occurrences within the stimuli to perform above chance. Therefore, if a version of the experiment could present the stimuli in a Latin script, this might ease all participants' cognitive load and allow for better performance.

Experiment 3

Presenting the characters in the BACS-1 font may have expanded the horizon of participant recruitment, but the unfamiliarity of the characters may have required additional effort to process and learn. This took away cognitive resources from the learning of higher-level statistics, and prevented observable co-occurrence learning to occur. Ideally, the authors' aim would be to develop a version of this paradigm with mean 2M accuracy above chance in order to see more meaningful variability in scores. One further drawback from the stimuli in previous versions of the experiment was the choice to create stimuli for the two languages which were only partially overlapping in the characters used to form them. This was selected to make stimuli ecological and similar to European languages, for example: most of these use the standard Latin alphabet, with some language-specific additions (“é”, “ô”, “ß”, “œ”). However, this gives participants some indication as to which language morphemes belong to, beyond pure co-occurrence statistics as intended.

For these reasons, the experiment was adjusted to present stimuli in Latin characters fully overlapping across both languages.

Methods

Participants

In order to present stimuli in Latin characters while retaining the novelty of the strings presented to participants, the range of languages spoken by participants had to be limited. A set of languages were chosen to keep the variety of language backgrounds as wide as possible. However, researchers had to ensure that a string presented in the experiment was not a real lexical item in any of the languages spoken by participants. To this end, the chosen languages had to be similar to each other to prevent the exclusion of too many random strings that might be used in the experiment. These languages were selected as English, German, Dutch, Spanish, Portuguese, French, Catalan, and Italian.

Therefore, the participant sample size and criteria remained identical to the previous two experiments, with the new restriction of them speaking only the languages previously mentioned, or any other language not using a Latin script.

Stimuli

Given the necessity to generate morpheme strings that are not real lexical items in any of the above listed languages, a pipeline was developed for their generation. All possible 3- to 5-character combinations of all the letter of the Latin alphabet were assembled, and any item present in these languages was excluded. From the remaining sample, 600 were randomly selected, and replaced by new random choices if any are at a Levenshtein distance of less than 2 or have 2-character chunks within them in common with another string. This leaves a pool of 572 strings, from which each participant gets a random selection of the 30 morphemes to be used in their submission. The result of this new stimulus generation procedure was that the morphemes for both language 1 and language 2 contain all (unaccented) letters of the Latin alphabet, and are therefore completely indistinguishable at the letter level.

Procedure

Given that Latin characters are easier to process than BACS-1 characters, this may lead to an over-familiarity with the presented morphemes, thereby causing participants to select all 2M combinations as congruent. A large “yes” bias would greatly diminish participants’ ability to reject between-language combinations. Indeed, a pilot experiment with 10 participants revealed that the c value of participants’ responses to the 2M condition was -0.72. To correct this over-familiarity, the morpheme familiarity test might be removed. A pilot was done under these conditions, and yielded a much more acceptable c of -0.34, reassuringly in line with previous experiments in this paper (experiment 1 c = -0.37; experiment 1 c = -0.35). The mean familiarity score for this pilot was also 57%, in line with both previous versions. This choice was therefore chosen as giving researchers the closest match to previous versions and their “yes” biases, and offered the best chance for variability in responses. Experiment 3’s procedure was therefore identical to experiment 1.

Analysis

Data analysis was identical to experiment 2.

Results

Participants

Two hundred participants were recruited for the third experiment. Four participants were then found to have average response times during testing over 2 seconds, which led to their exclusion. One participant responded “yes” to all items presented in the testing phase, and was also excluded. This left 195 participants’ data in the final analysis sample. These were made up of 138 males, and participants’ mean age was 22.21 years.

Group-level analyses

Familiarity test

Despite the removal of the morpheme spelling task, familiarity scores rose considerably ($t(194) = 17.91$, $p < 2.2e^{-16}$, estimate = 66%). This represents a 9% increase from experiment 2. The mean response time was also approximately 150 ms quicker in this version, at 1515 ms on average. This increase in accuracy combined with a decrease in response times provides evidence that participants segmented the words more easily than previously.

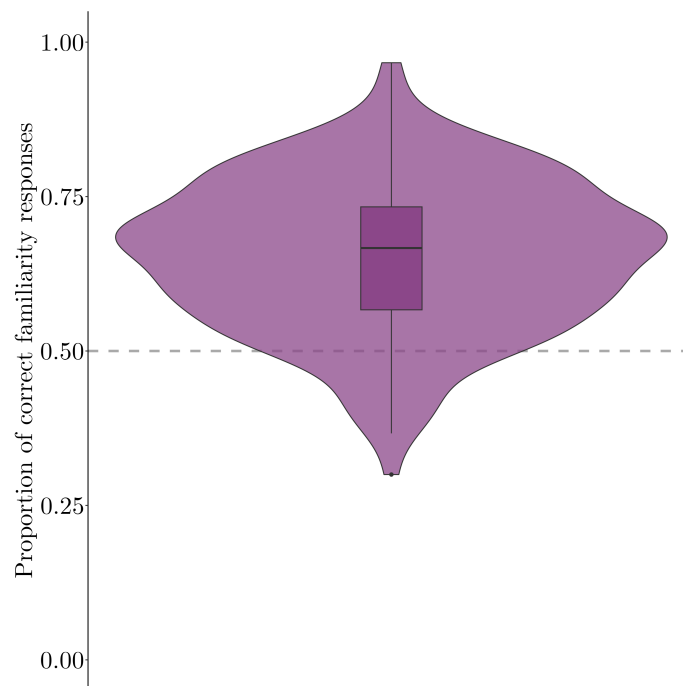


Figure 21: Violin plot and boxplot of the accuracy of familiarity test responses. The dashed line indicates chance level.

Morpheme familiarity effect

In line with both previous experiments, the morpheme familiarity effect was strong also in experiment 3. Linear modelling found significant differences in responses between the 0M (Intercept, $\beta = -0.50$, $z\text{-value} = -10.96$, $p = 6.2e^{-28}$), 1M ($\beta = 0.63$, $z\text{-value} = 18.16$, $p = 1.0e^{-73}$), and 2M ($\beta = 1.25$, $z\text{-value} = 35.22$, $p = 1.0e^{-271}$) conditions. Estimated marginal means revealed that participants tended to reject items from the 0M condition (model fit of the proportion of “yes” responses = 0.38, $SE = 0.01$), responded near but above chance level for 1M items (model fit = 0.53, $SE = 0.01$), and strongly preferred 2M items (model fit = 0.68, $SE = 0.01$). This is a pattern of responses more balanced around chance level: while in experiment 2, responses to the 0M condition were those around chance level, participants in this design were more able to respond “no” to 0M items, and we see instead the cluster of 1M responses fall around chance. This might indicate some hesitation for participants: it seems clearer to reject items with no familiar morphemes here, but those with one familiar word part tied to an unfamiliar string elicit some confusion as to whether to accept these.

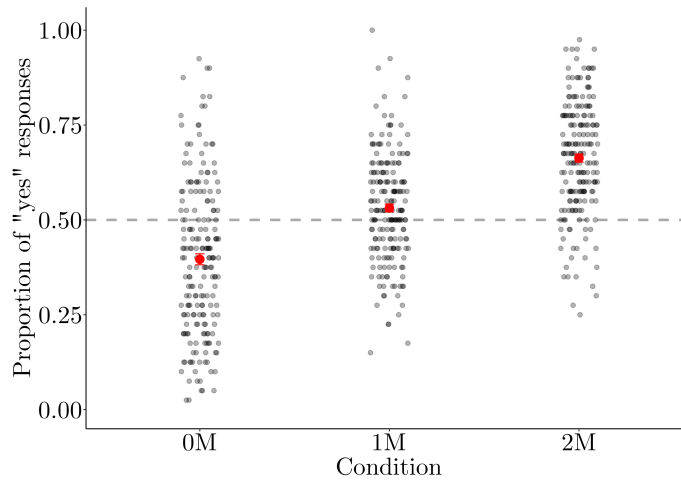


Figure 22: Proportion of “yes” responses in each testing condition. The dashed line indicates chance level. The red circle shows the group mean, with bars as the standard error of the mean.

2M performance

As for performance in the 2M condition, participants were able to distinguish between within- and between-language combinations ($t(194) = 2.68$, $p = 0.008$, estimate = 0.51). The mean $d\text{-prime}$ of participants was 0.07 ($t(194) = 2.61$, $p = 0.01$), indicating further that the underlying pattern was detected. The mean c value was of -0.44 ($t(194) = -14.40$, $p < 2.2e^{-16}$). This is a slightly stronger “yes” bias than in previous versions, but with responses more in tune with stimulus nature.

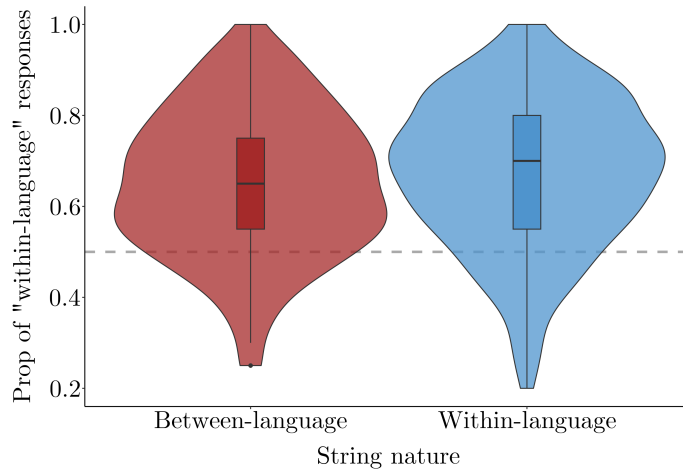


Figure 23: Boxplots of the proportion of “within-language” responses to each 2M combination type. The dashed line indicates chance level.

Individual differences analyses

BLP

Analysis of the Bilingual Language Profile showed 23 participants only speaking one language, 91 speaking two, 55 speaking three, and 26 speaking four. The most common L1 was Arabic, English the most common L2, and French the most common L3 and L4. Of the declared languages, these belonged to 6 different language families, and represented 13 different language branches. This sample therefore had 105 participants who used languages with one same script, and 90 who used languages with different scripts.

PCA

The Principal Components Analysis on experiment 3’s BLP data revealed four components of interest. PC1 loaded L4 history, use, proficiency, and attitudes, and accounted for 21% of variance in the data. PC3 was the equivalent for L3 metrics, explaining 20% of variance. PC2, like in experiments 1 and 2, looked at the balance between L1 and L2 language use. This was responsible for 10% of variance. Finally, PC8, also present in experiments 1 and 2, included L2 history information, and held 6% of variance.

Familiarity test

In this version of the experiment, none of our measures significantly predicted familiarity responses. It is possible that stimulus parameters (namely, their presentation in a familiar script) made segmentation so straightforward as to not require the aid or involvement of

other linguistic capabilities.

Morpheme familiarity effect

Overall, participants who had more balanced multilingual profiles (as computed by cosine similarity) tended to respond with a stronger “yes” bias than other participants (cosine similarity main effect $\beta = 3.33$, $z = 2.52$, $p = 0.01$).

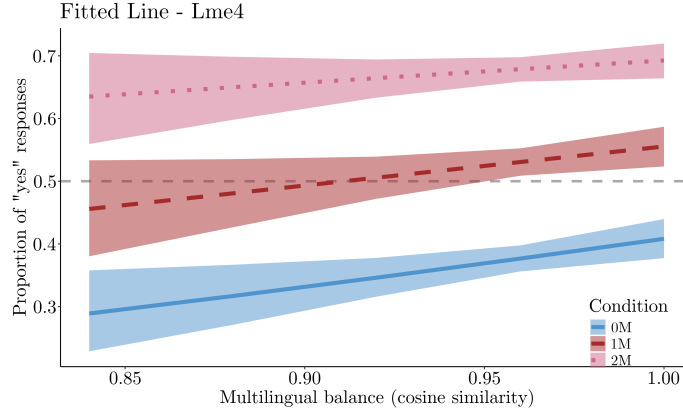


Figure 24: Model estimates of the effect of the amount of “yes” responses over all testing conditions, in relation to multilingual balance (cosine similarity). Cosine similarity scores closer to 1 indicate more balance across a participants’ languages. Shaded areas around the lines indicate the 95% confidence interval. The dashed line at 50% indicates chance level.

As in experiment 2, participant gender influenced propensity to respond “yes” to known morphemes, with women showing not only a lesser proportion of “yes” responses over all items (Women main effect $\beta = -0.25$, $z = -2.45$, $p = 0.01$), but also different responses across stimuli than men (1M*Women interaction $\beta = 0.15$, $z = 1.98$, $p = 0.048$; 2M*Women interaction $\beta = 0.18$, $z = 2.29$, $p = 0.02$).

Multiple multilingual metrics also affected “yes” responses. PC1, loading L4 information, revealed accentuated differences between the different testing conditions when participants had low L4 scores (2M*PC1 interaction $\beta = -0.11$, $z = -3.04$, $p = 0.002$). Another PC component, PC2, showed an effect of increase second-language: the more participants made use of their L2 rather than their L1, the stronger their morpheme familiarity effect was (1M*PC2 interaction $\beta = 0.10$, $z = 2.88$, $p = 0.004$; 2M*PC2 interaction $\beta = 0.19$, $z = 5.24$, $p = 1.61e^{-7}$). Higher levels of multilingual balance (entropy) also exaggerate the morpheme familiarity effect (1M*entropy interaction $\beta = 0.18$, $z = 2.70$, $p = 0.01$; 2M*entropy interaction $\beta = 0.28$, $z = 4.02$, $p = 5.20e^{-5}$).

Further metrics also interacted with the different testing conditions. Participants with

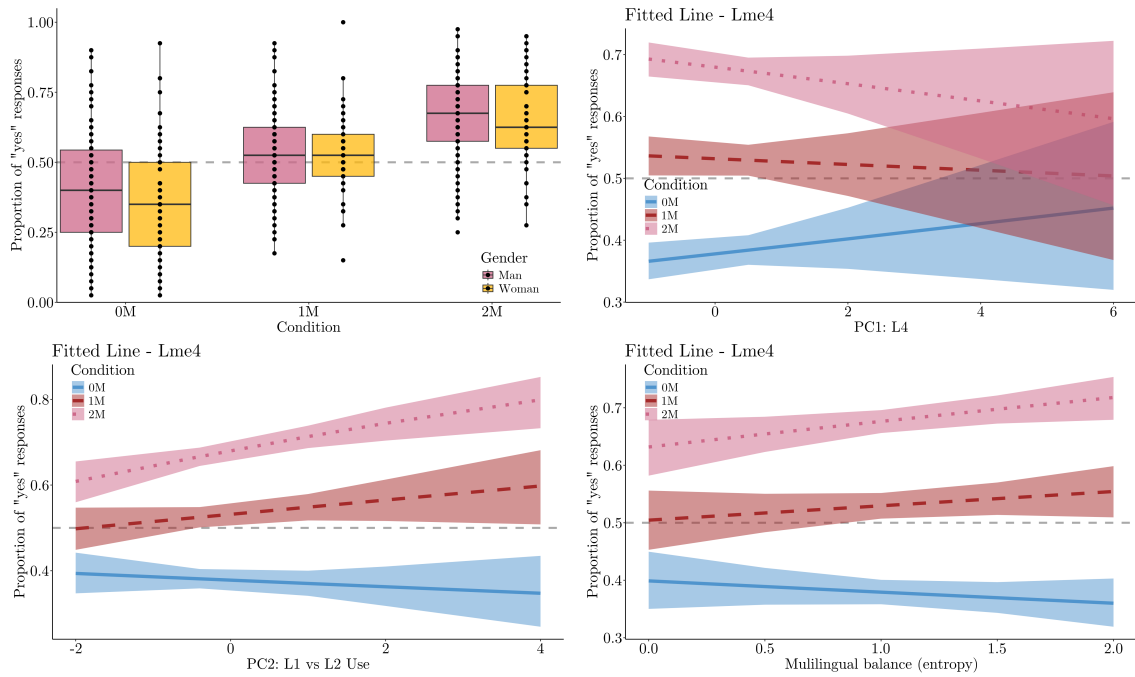


Figure 25: Model estimates of the effect of gender (top left), PC1 (L4; top right), PC2 (L1 vs L2 use; bottom left), and multilingual balance (entropy; bottom right) on "yes" responses, for the 0M (blue; solid), 1M (red; dashed), and 2M (pink; dotted) conditions. Higher levels of PC2 means more L2 rather than L1 use. Higher levels of entropy mean a more balanced multilingual profile. Shaded areas around the lines indicate the 95% confidence interval. The black dashed line at 0.5 indicates chance level.

more multilingual experience were more sensitive to the presence of familiar morphemes in testing input (2M*multilingual experience interaction $\beta = 0.10$, $z = 2.71$, $p = 0.007$). Increased balance between participants' first languages also boosted morpheme sensitivity (1M*L1-L2 difference interaction $\beta = -0.10$, $z = -2.77$, $p = 0.006$; 2M*L1-L2 difference interaction $\beta = -0.15$, $z = -4.19$, $p = 2.85e^{-5}$). There also seems to be a different pattern of morpheme awareness depending on the number of languages that participants declared to speak. Bilinguals (2M*bilinguals interaction $\beta = 0.24$, $z = 2.07$, $p = 0.04$) and trilinguals (1M*trilinguals interaction $\beta = 0.36$, $z = 2.07$, $p = 0.04$; 2M*trilinguals interaction $\beta = 0.54$, $z = 4.38$, $p = 1.17e^{-5}$) both have a stronger morpheme familiarity effect than monolinguals. Quadrilinguals, on the other hand, have different amounts of “yes” responses only for the 2M condition (2M*quadrilinguals interaction $\beta = 0.32$, $z = 2.25$, $p = 0.02$).

“yes” responses across conditions				
	β	SE	z-value	p-value
Cosine similarity	3.30	1.26	2.61	0.009
Women	-0.25	0.10	-2.45	0.01
Women*1M	0.15	0.08	1.98	0.047
Women*2M	0.18	0.08	2.29	0.02
PC1 (L4)*2M	-0.11	0.04	-3.05	0.002
PC2 (L1 vs L2 use)*1M	0.10	0.03	2.89	0.004
PC2 (L1 vs L2 use)*2M	0.19	0.04	5.25	$1.5e^{-7}$
Entropy*1M	0.18	0.07	2.71	0.007
Entropy*2M	0.28	0.07	4.04	0.00005
Multilingual experience*2M	0.10	0.04	2.71	0.007
L1-L2 difference*1M	-0.10	0.03	-2.77	0.006
L1-L2 difference*2M	-0.15	0.04	-4.20	0.00003
Bilinguals*2M	0.24	0.11	2.07	0.04
Trilinguals*1M	0.36	0.12	3.01	0.003
Trilinguals*2M	0.54	0.12	4.39	0.00001
Quadrilinguals*2M	0.32	0.14	2.25	0.02

Table 4: Summary of experiment 3 mixed modelling statistics of “yes” responses across conditions, for significant variables.

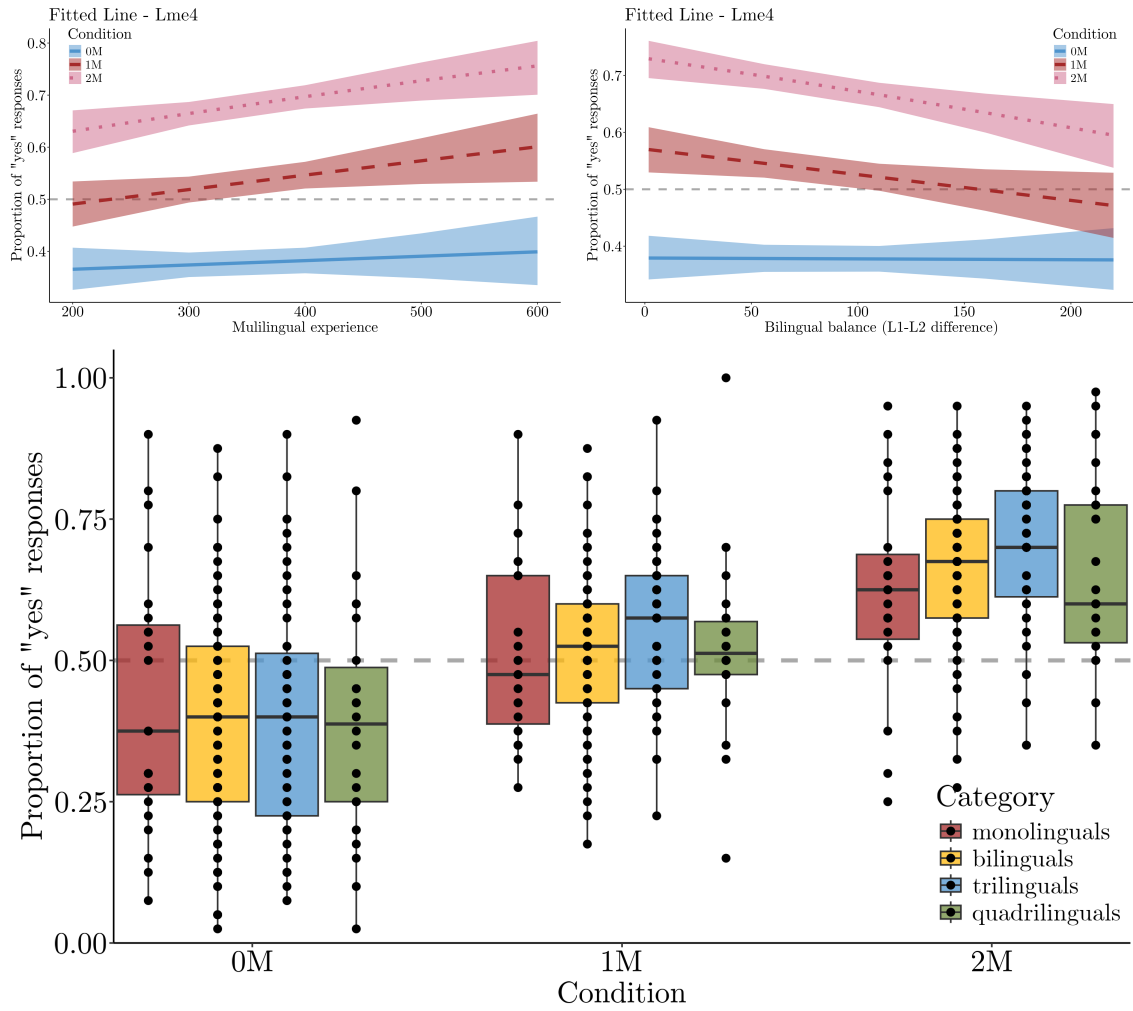


Figure 26: Model estimates of the effect of multilingual experience (top left), bilingual balance (L1-L2 difference; top right), and multilingual category (bottom) on “yes” responses, for the 0M (blue; solid), 1M (red; dashed), and 2M (pink; dotted) conditions. L1-L2 difference scores closer to 0 indicate a more balanced bilingual profile. Shaded areas around the lines indicate the 95% confidence interval. The black dashed line at 0.5 indicates chance level.

2M performance

Isolating focus on the key 2M condition, there were some main effects on “yes” responses in that condition. More L2 use compared to L1 use led to more “yes” responses in the 2M condition (PC2 main effect $\beta = 0.18$, $z = 3.10$, $p = 0.002$). These responses were also increased for participants with more multilingual experience (multilingual experience main effect $\beta = 0.15$, $z = 2.59$, $p = 0.0098$). Those with more balance, either bilingual or multilingual, responded “yes” more as well (L1-L2 difference main effect $\beta = -0.15$, $z = -2.74$, $p = 0.006$; cosine similarity main effect $\beta = 0.11$, $z = 1.99$, $p = 0.047$). In this version of the experiment, participants who spoke language with different scripts also tended to respond “yes” more often in the 2M condition (script main effect $\beta = 0.24$, $z = 2.14$, $p = 0.03$).

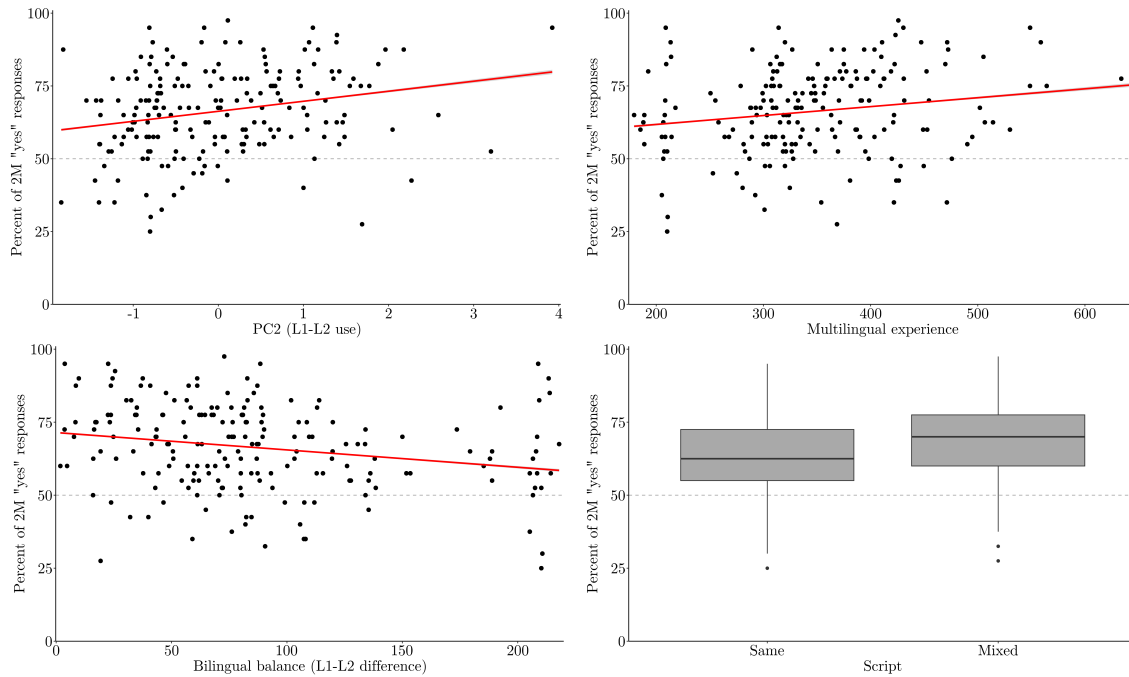


Figure 27: Plots of the percent of “yes” responses in the 2M condition as a function of PC2 (L2 vs L1 use; top left), multilingual experience (top right), bilingual balance (L1-L2 difference; bottom left), and script type (bottom right). Higher levels of PC2 means more L2 rather than L1 use. Lower levels of L1-L2 difference indicate more balanced profiles. The dashed line at 50% indicate chance level.

Crucially, three variables also predicted 2M accuracy. Most interesting given the results from the previous experiment, multilingual balance (cosine similarity) was once again a significant predictor of 2M accuracy (cosine similarity interaction $\beta = -0.13$, $z = -2.56$, $p = 0.01$). However, in this version of the experiment, this effect preserved its significance when calculated without monolingual participants (cosine similarity interaction $\beta = -0.11$, $z =$

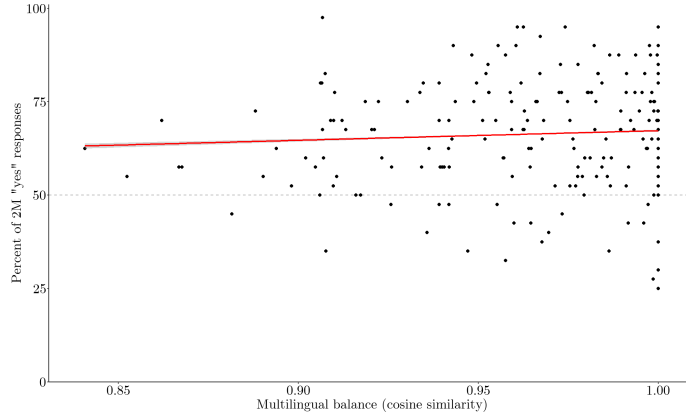


Figure 28: Plot of the effect of multilingual balance (cosine similarity) on 2M “yes” responses. Cosine similarity scores closer to 1 indicate more balance across a participants’ languages. The dashed line at 50% indicates chance level.

-2.02, $p = 0.04$). This relationship was, however, in an unexpected direction: more balanced multilinguals were less accurate. Exploratory analyses found the same pattern of results when calculating cosine similarity using language use scores as opposed to overall language scores, as is sometimes done in past literature (use cosine similarity $\beta = -0.11$, $z = -2.24$, $p = 0.03$). This would indicate that the previously detailed cosine similarity effect might be influenced in this direction through its language use components.

Finally, the bilingual group had significantly higher accuracy than monolinguals (bilingual interaction $\beta = 0.34$, $z = 2.06$, $p = 0.04$). In order to ensure that this group’s accuracy was also significantly above chance level instead of the linear model intercept, a post-hoc Student’s t -test was conducted. This confirmed that this group performed above chance level ($t(90) = 3.46$, $p = 0.0008$, estimate = 0.52).

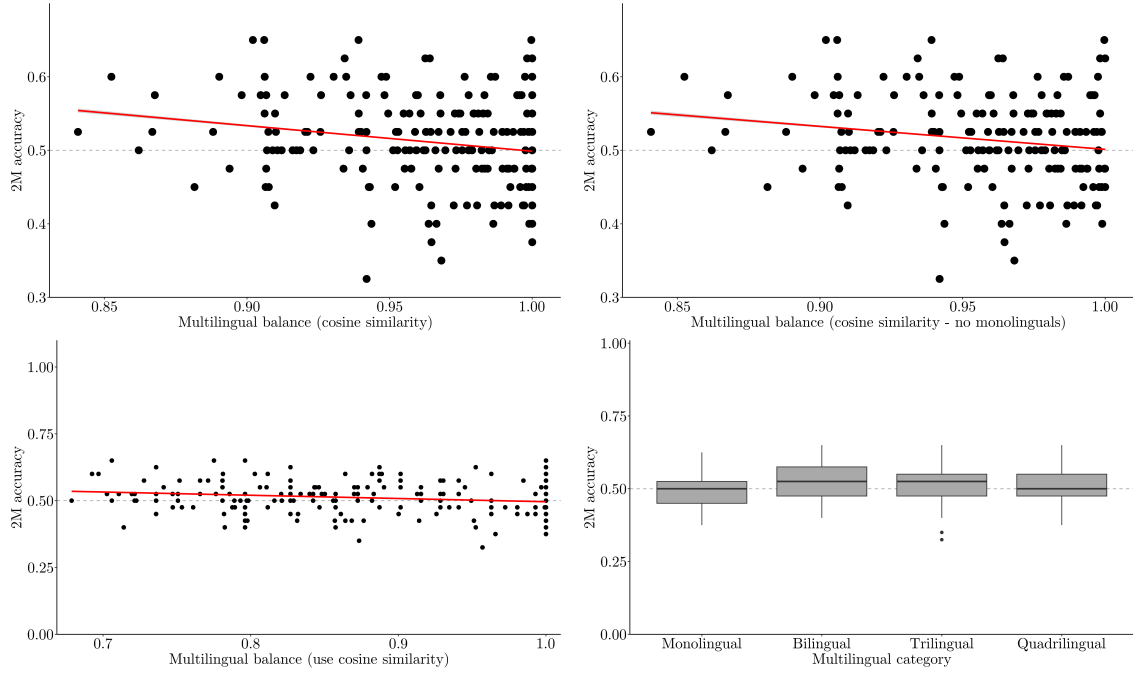


Figure 29: Plots of the relationship between accuracy in the 2M condition and multilingual balance (cosine similarity, top left; without monolinguals, top right; use cosine similarity, bottom left) and multilingual category (bottom right). The dashed line at 0.5 indicates chance level.

2M “yes” responses and accuracy				
	β	SE	z-value	p-value
PC2 (L1 vs L2 use)	0.18	0.06	3.10	0.002
Multilingual experience	0.15	0.06	2.58	0.0098
L1-L2 difference	-0.15	0.06	-2.74	0.006
Mixed script	0.24	0.11	2.14	0.03
Cosine similarity	0.11	0.06	1.99	0.047
Cosine similarity*correct	-0.13	0.05	-2.56	0.01
Cosine similarity (no monolinguals)*correct	-0.11	0.05	-2.02	0.04
Use cosine similarity*correct	-0.11	0.05	-2.24	0.03
Bilinguals*correct	0.34	0.16	2.08	0.04

Table 5: Summary of experiment 3 mixed modelling statistics of 2M “yes” responses and accuracy across conditions, for significant variables.

Discussion

Several design modifications aimed to make this third version of the experiment easier for participants. The conversion of BACS-1 stimuli into Latin characters and the consequent removal of the morpheme spelling task certainly succeeded in increasing word segmentation ability, as this was reflected in both higher accuracy and faster reaction times in the morpheme familiarity task. These modifications seem to have made the stimuli simpler to process. That no multilingual metrics impacted segmentation ability might indicate that the stimuli might have been learned sufficiently well, with the main brunt of cognitive processing being shifted to the tracking of other regularities. Segmentation at this level of difficulty may not require different cognitive strategies developed from multilingualism.

A higher probability of classifying stimuli as congruent arose for the male participants, and for those multilinguals who were more balanced across all of their languages. Gender differences as a main effect were only found in this last version of the experiment, and fall outside the scope of this study, and will not be discussed further. On the other hand, participants engaged in regular switching and/or a more egalitarian view of their languages might have more fluid cross-linguistic transfers, which might lead them to accept a wider range of stimuli as linguistic.

The morpheme familiarity effect at group-level saw a downward shift compared to the previous versions of the experiment. The average proportion of “yes” responses in the 0M were at 38%, with the 1M condition taking its place at 53%. Responses to the 2M condition were far above this at 68%. This could be due to the aforementioned change in stimulus nature, and the consequent rise in familiarity. With a better grasp on which morphemes they saw in training, participants were better able to recognise and reject strings that didn’t contain any of these familiar morphemes. As task instructions remained deliberately vague (“do these words belong to what you saw previously?”), participants could interpret the 1M combinations either as incongruent due to the presence of an unknown word part, or congruent due to the presence of a familiar morpheme. In this experiment, participants seems approximately balanced in their preferences for either interpretation.

Women deviated from their slightly more negative evaluation of strings for this 1M condition, possibly due to this ambiguity impacting both genders equally. As for the multilingual measures, the morpheme familiarity effect was strengthened for participants who used their L2 more than their L1 (PC2) and were more balanced across their first top language (L1-L2 difference). Bilinguals were also more sensitive than their monolingual

counterparts. This would seem to indicate a kind of bilingual advantage, with language use and balance increasing morphological awareness. This is extended by the result that multilingual experience and multilingual balance (entropy) also lead to a larger morpheme familiarity effect. However, the picture gets blurrier when adding multilingual metrics beyond two languages. Higher L4 scores (PC1) *reduce* morpheme familiarity effects, and quadrilinguals are the multilingual group with the weakest effect. Indeed, looking at the different multilingual categories, quadrilinguals have less variability in “yes” responses based on testing condition, followed by the monolinguals, then bilinguals, with trilinguals seeing the strongest effect. This pattern is therefore that of an advantage up to the trilingual level, and then a cost of adding a fourth language. Given the youth of participants (18-25 years age range, 22.21 years on average), having a fourth language in such a short timespan could entail lessened or shallower acquisition of all of a participant’s languages due to having to dedicate time to a fourth. This theory would certainly also depend on age of acquisition and use habits of a multilingual’s languages. However, since PC1 is sensitive to within-group differences, this analysis also indicates that those quadrilinguals with stronger L4 history/use/proficiency/attitudes have an even larger cost than those with a lesser impact of a fourth language. The pattern of results may also be impacted by the lower statistical power in the quadrilingual group (only 26 quadrilinguals out of 195 participants), but is a peculiar finding that warrants future research.

In the 2M condition, the bilingual metrics of increased L2 compared to L1 use and bilingual balance led to even more sensitivity to the combined presence of two known morphemes, regardless of the combinatorial structure. This reflects the generalised sensitivity discussed above. Multilingual experience also similarly increases “yes” responses in the 2M condition. Interestingly, participants with knowledge of languages using more than one script also accept two-morpheme combinations, but this effect is separate from an overarching morpheme familiarity effect. Experience with manipulating linguistic systems with fewer visual differences could enhance visual processing abilities when the stimuli represent a very likely lexical candidate.

As to whether the stimuli changes freed up enough cognitive resources from the necessary segmentation to allow for more successful co-occurrence tracking, the group-level results do show above-chance performance in the discrimination of within- and between-language morpheme pairings. This is a very small group effect (1% above chance level), but this was expected by researchers (see experiment 1 section effect size motivations). This is nonetheless significantly above chance in a very novel implicit paradigm, and may represent

meaningful, if slight, co-occurrence tracking and categorisation abilities by participants on a group level. This difference was found despite a stronger c bias in the 2M conditions than in previous experimental versions, which is due to the stronger morpheme familiarity effect detailed above and the change in stimulus script. This shows that categorical boundaries can indeed be learned implicitly through repeated exposure to words with co-occurrence regularities.

Even more interestingly, the accuracy with which participants were able to track these co-occurrences depended on some of the same factors that dictated their sensitivity to the presence of the constituent morphemes. Bilingual participants were the only group to perform above chance level, indicating once more the possibility of a bilingual-specific advantage. Additionally, we find another cost at the quadrilingual level in terms of multilingual balance (cosine similarity). Further information from the balance of multilingual use (use cosine similarity) intimates that this disadvantage may be the consequence of exaggerated *use of a fourth language*. Whereas the PCA in the previous version of the experiment did pick up L4 Use as a valuable component, it was not so in this version, and therefore was not examined in isolation. However, information on the presence and use of a fourth language certainly seems to play a crucial role in mediating the bilingual (and in some case, trilingual) benefit observed here in terms of increased awareness and co-occurrence tracking abilities of morphemes.

General Discussion

Creation of a novel implicit bilingual SL task

The present project followed the development of an implicit bilingual statistical learning task, in which the artificial words presented to participants were formed of morphemes that co-occurred only with other morphemes of that language. The task for participants was to segment these words to learn the morphemes, and track these co-occurrence patterns to be able to accurately reject between-language combinations presented in the testing phase. This two-pronged task therefore called for both extraction and integration using the frequency-based and co-occurrence statistical information present in the artificial stimuli.

This paradigm allowed for the replication of the morpheme familiarity effect found in [Lelonkiewicz, Ktori, et al. \(2020\)](#). Participants were able to develop a sensitivity to morphemes in novel test words following only a passive word exposure phase. This uptick in their feeling of congruency going from test strings with no known morphemes to those with one morpheme present, to strings composed of two known morphemes was a very solid phenomenon observed across all three experiments. This was the case even for the first experiment, in which the design made the task the most difficult.

Interestingly, the experiment 3 stimuli font change to Latin characters had the consequence of reducing the proportion of “yes” responses across all conditions. The one-morpheme items received affirmative evaluations half of the time, which nicely reflects their two possible interpretations. The presence of a foreign element within the test string may cause for its rejection, or the recognition of a familiar element might be enough for the participant to generalise the learned patterns and accept a novel pairing, much as an English learner knowing “read|er” might accept “read|ing”.

The morpheme familiarity effect was reinforced based on multilingual factors. Participants who made more use of their second rather than their first language had greater separation of their responses to the different testing conditions (in experiments 2 and 3). The balanced use of an L1 and L2, or the greater use of an L2 over an L1 would be more cognitively effortful to allow for communication in the weaker language. However, over time, this may have led to greater linguistic metacognitive processing as they become more bilingual ([Ordin et al., 2020](#)). This was also the case for those with more bilingual balance (more similar scores for their L1 and L2; experiments 2 and 3). As it has been proposed that bilingual balance has links to language processing abilities ([Birdsong, 2015](#)), these

participants may be more efficient at morphemic identification. This evidence would support the theory that those with more bilingual (as opposed to monolingual) profiles have built up greater sensitivity to familiar chunks. Additionally, in experiments 1 and 3, trilinguals were the group with the best morpheme familiarity, followed by bilinguals, with quadrilinguals' performance falling below even that of monolinguals. This would indicate that there is at least a bilingual morphemic sensitivity effect that may be reversed beyond 3 languages.

Within the two-morpheme condition, the experimental conditions in the final version were sufficient to find a significant effect of implicit bilingual co-occurrence learning even at a group level. Participants were therefore able to track co-occurrences across time and the different training blocks in which the morphemes were presented to infer that this web of pairings denotes items coming from different languages. These statistical regularities learned completely unconsciously were sufficient to create mental categorisation of the input.

While the size of the effect is very modest, this is an ability that was significantly above chance level. This design is both novel and complex, which may have contributed to the difficulty in finding more decisive results. Indeed, participants are required to first engage in extractive statistical learning to find the morphemes within the words, and successively integrate the morphemes into categories thanks to co-occurrence tracking mechanisms. There may be ways to facilitate the segmentation, but the integration seems to be the trickier process to manipulate. None of the multilingual metrics reliably predicted familiarity scores across multiple experimental versions, and none did so in experiment 3. This may have been anticipated, as [Poepssel et al. \(2016\)](#) suggested that a bilingual advantage may only exist for integrative statistical learning.

The question remains as to how to make the integration process easier, while keeping the statistical learning manipulation implicit. More cognitive resources might simply be freed up for integration thanks to conditions supporting easier segmentation, but the unknowns behind the process of co-occurrence itself limit the extent to which this can be directly observed. The statistical regularities are already fully non-overlapping across languages (that is, a morpheme from language 1 was never shown with one from language 2 during training), so this categorisation cue was fully reliable. Other experimental manipulations to make this cue more salient unfortunately sacrifice the desired implicitness of the training.

However, the interpretation of this implicit co-occurrence structure by participants may not have been the one conceived of by experimenters during the design process. Despite the lack of biasing instructions favouring the interpretation that the input was monolingual,

all of the testing words may have been learned as if they do come from a single language. Consider the below table of mock stimuli in English:

	“L1”	“L2”
congruent	sting - er	walk - ing
congruent	weak - est	quick - ly
“incongruent”	weak - ly	walk - er
“incongruent”	sting - ing	quick - est

In this “L1”, two stems are associated with “-er”, a suffix indicating a noun part of speech, and “-est”, which indicates an adjective or adverb. “L2” has the suffixes “-ing”, for verbs, and “-ly”, for adverbs. However, from looking at the “incongruent” cross-language combinations, it is perfectly plausible for one stem to be associated with different suffixes, and even unusual combinations of known stems and suffixes would not necessarily be cause for rejection (“giraffing” could be a form of animal play-acting similar to monkeying). Therefore, instead of treating a “between-language” combination as a violation, it could be understood by a similar explanation as a form of generalisation of learned morpheme part of speech indicators. However, even if this was the case in the present experiment, this would still require the same mental integration of classifying linguistic stimuli as different parts of speech, and would still involve co-occurrence tracking.

Multilingual variables

One of the strengths of the present experiment was the ability to recruit participants from a very wide variety of language backgrounds. Through the use of artificial stimuli, and especially with the BACS-1 characters from the first two experiments, the authors were able to include participants from any geographical region and speaking any languages, with the only requirement being English fluency to understand task instructions. The findings outlined here are therefore not restricted to specific stimulus language combinations or to specific bilingual participant profiles. This allows for more generalisability of the statistical learning effects found. Wide-ranging artificial statistical learning studies may be a way to overcome the anglocentrism and eurocentrism that characterise most studies today.

Several conclusions can therefore be drawn about the effect of multilingual variables in general with relation to statistical learning in this task. The only variable that predicted

co-occurrence tracking ability was multilingual balance, as calculated by cosine similarity. The more balanced the multilingual profile across up to four spoken languages, the worse they seemed to be at forming these statistical regularity-based categories. Past research had found a role for increased *bilingual* balance providing a benefit in a different bilingual statistical learning paradigm (Onnis et al., 2018). However, there was no effect of bilingual balance in this experiment, meaning that the cause here must have multilingual roots in effects beyond two languages. Additionally, there was no effect on multilingual experience on co-occurrence tracking: the mere accumulation of linguistic knowledge did not play a role. Looking across all versions of the experiment, the bi-/tri-/quadrilingual categories did not have a consistent effect either; the cosine similarity significance therefore cannot be attributed to multilingual status, but seems to have to do with how third and fourth languages are being learned, used, and valued.

The participants' age range was set as 18-25 in order to remove any age-related statistical learning differences. However, being perfectly proficient or balanced in four languages at that young is perhaps a difficult thing to be: having equal capacities in four languages at that age might mean shallower learning of all (or most) of them. This phenomenon may lessen for older balanced multilinguals who have had more time to solidly learn all of their languages. An earlier acquisition, exaggerated use, or excessive valuing of a third and fourth language might have impacts on the ability to perform integrative statistical learning to form distinct categories. Gullifer and Titone (2020) posited that bilinguals are able to perform in a compartmentalised or integrated way. Additionally, Abutalebi et al. (2016) proposed that, according to the adaptive control hypothesis, the individual differences that change show bilinguals represent, access, and control their languages might determine how integrated their use is. It would stand to reason that the same can be extended to multilinguals. Those multilinguals who are more used to having their other languages co-activated in various settings may more easily integrate statistical regularities across languages.

As for the measure of cosine similarity more generally, it was initially developed to overcome the perceived mathematical correlation between entropy and multilingual experience. Empirically, a very high correlation was seen across all versions of the experiment: experiment 1 Pearson correlation $t(191) = 37.30$, $p < 2.2e^{-16}$, estimate = 0.94; experiment 2 Pearson correlation $t(182) = 30.18$, $p < 2.2e^{-16}$, estimate = 0.91; experiment 3 $t(193) = 32.95$, $p < 2.2e^{-16}$, estimate = 0.92. However, the correlation between cosine similarity scores and multilingual experience are only tentative: experiment 1 Pearson correlation $t(191) = -1.68$, $p = 0.09$, estimate = -0.12; experiment 2 Pearson correlation $t(182) = -1.62$,

$p = 0.11$, estimate = -0.12; experiment 3 Pearson correlation $t(193) = -1.46$, $p = 0.14$, estimate = -0.10. The correlation between entropy and cosine similarity also demonstrate this: experiment 1 Pearson correlation $t(191) = -3.60$, $p = 0.0004$, estimate = -0.25; experiment 2 Pearson correlation $t(182) = -4.85$, $p = 2.63e^{-6}$, estimate = -0.34; experiment 3 Pearson correlation $t(193) = -5.11$, $p = 7.87e^{-7}$, estimate = -0.34. Entropy therefore seems to be a measure at the crossroads between balance and multilingual experience, though being very largely biased by the latter. Cosine similarity is the more independent measure of the two.

However, some issues still remain surrounding the practicalities of its use. Given the calculation of balance either with global language scores or by conserving only proficiency or use, it is sometimes difficult to see patterns across literature. The calculation of cosine similarity can be undergone in the same way regardless of the choice made, but consensus on this point is greatly needed. Another point is that of possible cut-offs put in place in deciding which global language scores to accept in the calculation of cosine similarity. Participants were instructed to enter any number of languages they speak, but the choice to include or omit some very minimal understanding of one language may have important consequences on the calculated cosine similarity score. If a bilingual has the maximal score in their L1 and L2 (score vector = [218, 218]), they will get a cosine similarity of 1, representing perfect balance. However, if they elected to also mention their very dim knowledge of an L3 (score vector = [218, 218, 10]), their cosine similarity would fall to 0.83. On a scale from 0 to 1, this is a very significant difference. For this reason, the present authors chose as 50 the minimum global language score: this represents 23% of the perfect global language score, and we esteemed that this showed sufficient additional learning and knowledge to add to a multilingual's linguistic profile. However, this might be too strict, or too liberal; it is a first attempt to create a reasonable, representative measure of multilingual balance. The one other issue with this measure is the small variability: due to the nature of the global language scores, this number is necessarily positive, and restricted even more due to the cut-off. Across all experiments, the lowest cosine similarity score was 0.84. Another question can be asked of the monolinguals. By default, since they only speak one language, they have a cosine similarity of 1, but it cannot be said that this one-dimensional calculation represents balance. It would stand to reason to exclude monolinguals from cosine similarity analyses to preserve the true meaning of the measure.

In terms of what other multilingual measures might be particularly helpful in analyses, the Principal Components Analyses done often picked up similar components across all experiments. There were always L3 and L4 components, as well as L1 - L2 use and L2 history.

Their prevalence might suggest that these components are good bottom-up summaries of the multilingual experience. While L3 and L4 components have their natural place in research, L2 history (which charts answers to questions such as L2 age of acquisition) might explain more variance than previously thought. Theoretically, the length of time that one has juggled more than one language might be very significant in understanding the multilingual experience. The split between L1 and L2 use is also very theoretically compelling: the extent to which one stays in their dominant language or expends more cognitive effort while developing capabilities in a second language might be a component to preserve in future research.

Finally, the overall multilingual category that participants belong to does seem to be a useful catch-all to understand global effects. However, any general effect of being a bilingual or trilingual remains uninformative as to which aspect of bi- or trilingualism provoked said effects. It may be the starting point for further exploration into multilingual use, balance, or experience. However, more sensitive measures are definitely required to look specifically at language history, use, proficiency, attitudes, and balance in all of a participant’s languages to understand the complex dynamics of multilingualism.

Limitations

In terms of the experimental design, many elements may require tweaking to render the experiment more easily digestible to the statistical learner. Passive exposure during the training phase may not be engaging enough to sustain attention throughout the length of the training. This might also not be realistic, since real children interact with their caretakers and get feedback during learning (Frost et al., 2019). There may also not be enough repetitions of the stimuli during training to give the participant sufficient time to segment and track co-occurrences, especially in experiments 1 and 3, which did not have the morpheme spelling task to aid their segmentation. The number of individual morphemes used in the experiment might also be too high, which would complicate tracking. The modifications of experiment 3 did seem to make learning smoother for participants, but it is regrettable that this limited the participant backgrounds due to the necessity to validate the experimental stimuli.

The usage of 2-artificial forced-choice testing also only permits post-hoc statistical learning checks, not online learning measures to examine the timeline. It might have been interesting to see after which number of repetitions the morphemes were learned well enough to divert resources to co-occurrence tracking. Evidence of this learning might also be seen

earlier in neurological measures before they emerge in behaviour (Aguasvivas et al., 2024). It might also be possible through electrophysiological measures to detect larger differences in between- and within-language two-morpheme combinations. The statistics presented in statistical learning experiments are also very watered-down versions of the many regularities learners are bombarded with in their ordinary lives, and are rarely as neat (Frost et al., 2019). Real bilingual input would indeed also have both implicit and explicit cues to differentiate input from different languages (Weiss, Gerfen, et al., 2009).

Future directions

This statistical learning task is still in its infancy, and more marked effects might be seen thanks to some refinements. Further piloting could help identify the best parameters for the number of repetitions of the initial words in training, the number of distinct morphemes used, and so on. The 1M and 2M conditions might be removed to reduce the strong “yes” bias in the 2M condition and allow for participants to rely on finer instincts than the mere presence of two familiar morphemes. Additionally, due to the presence of BACS-1 characters in experiments 1 and 2, participants had no phonological information available to support their learning (Chetail, 2017). This was, to a lesser degree, also the case in experiment 3, where approximately half of the morphemes were just composed of consonants, and had low pronounceability (e.g. “VWNM”). These items were included to keep a large pool of morphemes for participants to be randomly assigned; in this way, having different morphemes for each iteration of the experiment would greatly reduce between-participant stimuli-related confounding factors. However, the morphemes with a vowel may be enough to satisfy this condition in a future experiment, and boost learning through the addition of phonology.

Conclusion

In the present study, a novel implicit bilingual statistical learning paradigm was trialled. The experimental design remains slightly too demanding for participants, but nonetheless revealed that bilingual morphemic learning can occur in the absence of explicit cues or instruction relating to the dual-language structure. A bilingual advantage was found for the detection of familiar morphemes in new input. While effect sizes were small, a possible quadrilingual disadvantage was observed, possibly due to participants processing input in a very integrated fashion instead of tracking co-occurrences within each stimulus language. The complexity of these effects, especially those arising when analysing participants past the bilingual level, demonstrates the need for refined, accurate measures of many different aspects of the multilingual experience.

Supplementary Materials

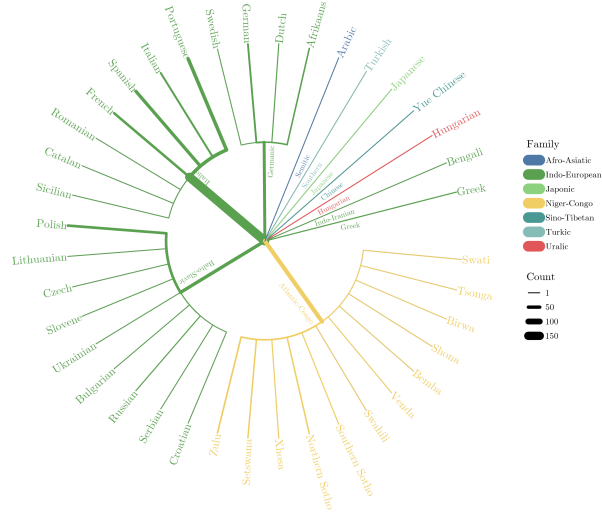


Figure 30: Tree plot showcasing the range of language backgrounds of participants in experiment 1. The colours indicate the different language families. The inner ring holds the various branches of those families, each of those ending in individual language on the outer ring. Line thickness indicates the density of each element.

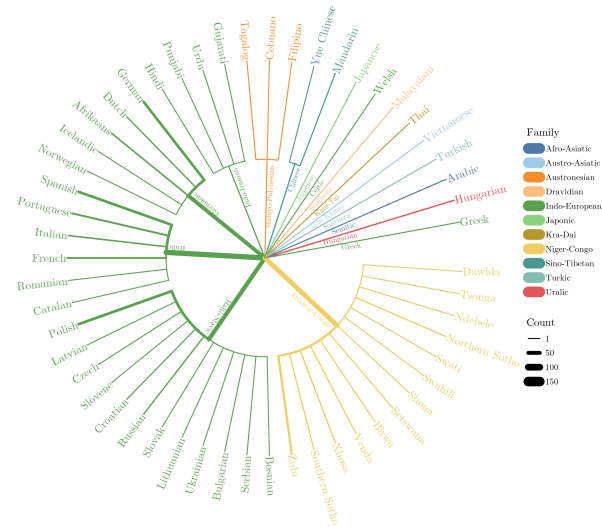


Figure 31: Tree plot showcasing the range of language backgrounds of participants in experiment 2. The colours indicate the different language families. The inner ring holds the various branches of those families, each of those ending in individual language on the outer ring. Line thickness indicates the density of each element.

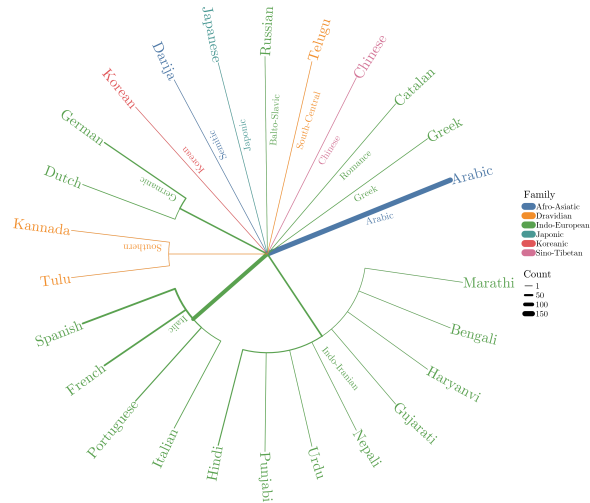


Figure 32: Tree plot showcasing the range of language backgrounds of participants in experiment 3. The colours indicate the different language families. The inner ring holds the various branches of those families, each of those ending in individual language on the outer ring. Line thickness indicates the density of each element.

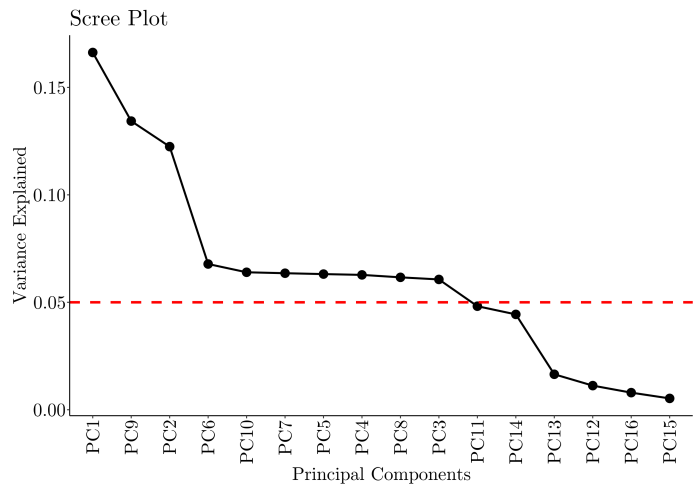


Figure 33: Scree plot of the PCA run on data for experiment 1. The red dashed line at 5% indicates the minimum variance explained by a component for it to be considered for analysis.

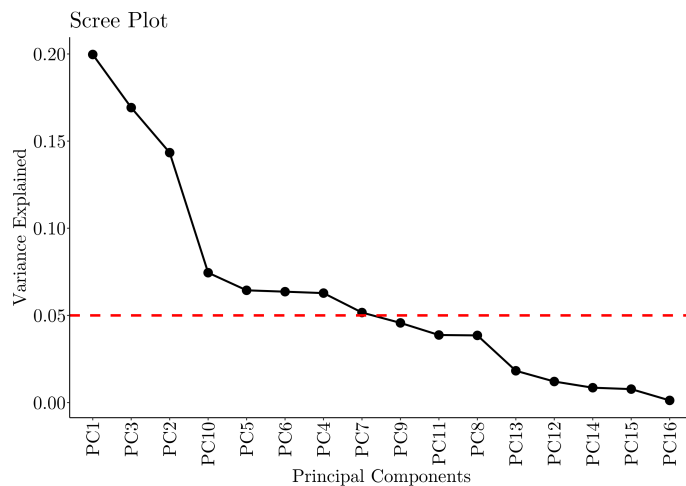


Figure 34: Scree plot of the PCA run on data for experiment 2. The red dashed line at 5% indicates the minimum variance explained by a component for it to be considered for analysis.

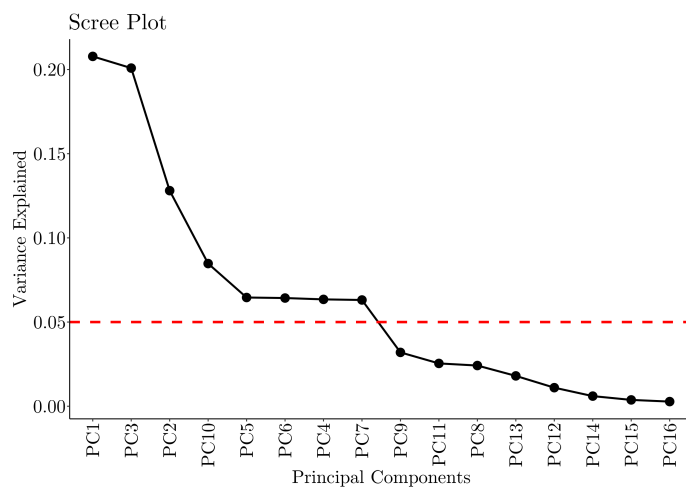


Figure 35: Scree plot of the PCA run on data for experiment 3. The red dashed line at 5% indicates the minimum variance explained by a component for it to be considered for analysis.

Exp. 1 Familiarity - Gender				
correct~scale(trialn)+Gender+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.04	6.95	3.7e-12 (***)
scale(trialn)	-0.06	0.03	-2.04	0.042 (*)
Other	0.48	0.25	1.91	0.057 (.)
Woman	-0.01	0.06	-0.22	0.82
Exp. 1 Familiarity - Age				
correct~scale(trialn)+scale(Age)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	9.17	4.6e-20 (***)
scale(trialn)	-0.06	0.03	-2.03	0.042 (*)
scale(Age)	0.06	0.03	1.91	0.056 (.)
Exp. 1 Familiarity - PC1 (L3)				
correct~scale(trialn)+RC1_L3+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	9.11	8.1e-20 (***)
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
PC1_L3	-0.03	0.03	-0.96	0.34
Exp. 1 Familiarity - PC9 (L4)				
correct~scale(trialn)+RC9_L4+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	9.1	9.0e-20 (***)
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
PC9_L4	0.02	0.03	0.67	0.5
Exp. 1 Familiarity - PC2 (L1 vs L2 Use)				
correct~scale(trialn)+RC2_use_L1vsL2+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	9.1	9.0e-20 (***)
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
PC2_L1vsL2_Use	-0.02	0.03	-0.65	0.52
Exp. 1 Familiarity - PC8 (L2 History)				
correct~scale(trialn)+RC8_hist_L2+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	9.11	8.5e-20 (***)
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
PC8_L2_History	-0.02	0.03	-0.83	0.41

Exp. 1 Familiarity - PC3 (L2 Proficiency)				
correct~scale(trialn)+scale(Age)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	9.17	4.6e-20 (***)
scale(trialn)	-0.06	0.03	-2.03	0.042 (*)
PC3.L2_Proficient	0.06	0.03	1.91	0.056 (.)
Exp. 1 Familiarity - Multilingual experience				
correct~scale(trialn)+scale(multiexp)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	9.1	9.4e-20 (***)
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
scale(Multilingual experience)	0.01	0.03	0.48	0.63
Exp. 1 Familiarity - Bilingual balance (L1-L2 difference)				
correct~scale(trialn)+scale(L1_L2_diff)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	9.11	8.3e-20 (***)
scale(trialn)	-0.06	0.03	-2.04	0.042 (*)
scale(L1-L2 difference)	-0.03	0.03	-0.92	0.36
Exp. 1 Familiarity - Multilingual balance (entropy)				
correct~scale(trialn)+ent+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.24	0.1	2.51	0.01 (*)
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
Entropy	0.03	0.07	0.36	0.72
Exp. 1 Familiarity - Multilingual balance (cosine similarity)				
correct~scale(trialn)+cossim+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.55	0.83	-0.66	0.51
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
Cosine similarity	0.86	0.87	0.99	0.32
Exp. 1 Familiarity - Script				
correct~scale(trialn)+script+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.27	0.03	8.44	3.2e-17 (***)
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
Different script	0.05	0.1	0.55	0.58

Exp. 1 Familiarity - Category				
correct~scale(trialn)+category+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.4	0.21	1.93	0.054 (.)
scale(trialn)	-0.06	0.03	-2.04	0.041 (*)
Bilingual	-0.14	0.21	-0.64	0.52
Trilingual	-0.17	0.22	-0.78	0.44
Quadrilingual	-0.09	0.22	-0.4	0.69

Exp. 1 ‘Yes’ responses - Conditions				
participant response~testing_condition+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.27	0.52		
1M	0.04	0.2	-0.19	
2M	0.27	0.52	-0.4	0.98
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.05	-1.06	0.29
1M	0.28	0.04	7.22	5.3e-13 (***)
2M	0.67	0.05	12.74	3.4e-37 (***)

Exp. 1 ‘Yes’ responses - Gender				
participant response~scale(trialn)+testing_condition*Gender+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.1	0.06	-1.71	0.09 (.)
scale(trialn)	-0.08	0.01	-5.17	2.3e-07 (***)
1M	0.28	0.05	5.99	2.1e-09 (***)
2M	0.69	0.05	14.43	3.2e-47 (***)
Other	0.09	0.35	0.26	0.8
Woman	0.11	0.09	1.2	0.23
1M*Other	0.51	0.29	1.72	0.09 (.)
2M*Other	0.77	0.33	2.33	0.02 (*)
1M*Woman	-0.02	0.07	-0.31	0.75
2M*Woman	-0.08	0.07	-1.02	0.31

Exp. 1 ‘Yes’ responses - Age				
participant response~scale(trialn)+testing_condition*scale(Age)+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.26	0.51		
1M	0.04	0.19	-0.13	
2M	0.24	0.49	-0.37	0.97
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.04	-1.13	0.26
scale(trialn)	-0.08	0.01	-5.18	2.2e-07 (***)
1M	0.28	0.04	7.27	3.6e-13 (***)
2M	0.68	0.05	13.07	4.9e-39 (***)
scale(Age)	-0.1	0.04	-2.21	0.03 (*)
1M*scale(Age)	0.07	0.04	1.92	0.054 (.)
2M*scale(Age)	0.14	0.05	2.77	5.6e-03 (**)

Exp. 1 ‘Yes’ responses - PC1 (L3)				
participant response~scale(trialn)+testing_condition*RC1_L3+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.04	-1.2	0.23
scale(trialn)	-0.08	0.01	-5.18	2.3e-07 (***)
1M	0.27	0.04	7.83	4.9e-15 (***)
2M	0.67	0.04	18.41	1.2e-75 (***)
PC1_L3	0.06	0.04	1.44	0.15
1M*PC1_L3	-0.03	0.04	-0.95	0.34
2M*PC1_L3	-0.05	0.04	-1.31	0.19

Exp. 1 ‘Yes’ responses - PC9 (L4)				
participant response~testing_condition*RC9_L4+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.27	0.52		
1M	0.04	0.2	-0.18	
2M	0.26	0.51	-0.4	0.97
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.04	-1.07	0.29
1M	0.28	0.04	7.23	4.9e-13 (***)
2M	0.67	0.05	12.77	2.5e-37 (***)
PC9_L4	-0.06	0.05	-1.33	0.18
1M*PC9_L4	0.03	0.04	0.78	0.44
2M*PC9_L4	0.03	0.05	0.61	0.54

Exp. 1 ‘Yes’ responses - PC2 (L1 vs L2 Use)				
participant response~scale(trialn)+testing_condition*RC2_use_L1vsL2+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.04	-1.22	0.22
scale(trialn)	-0.08	0.01	-5.19	2.2e-07 (***)
1M	0.27	0.04	7.83	4.7e-15 (***)
2M	0.67	0.04	18.43	7.2e-76 (***)
PC2_L1vsL2_Use	0.05	0.04	1.26	0.21
1M*PC2_L1vsL2_Use	0.02	0.04	0.61	0.54
2M*PC2_L1vsL2_Use	0.02	0.04	0.65	0.51

Exp. 1 ‘Yes’ responses - PC8 (L2 History)				
participant response~scale(trialn)+testing_condition*RC8_hist_L2+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.26	0.51		
1M	0.04	0.2	-0.22	
2M	0.27	0.52	-0.41	0.98
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.04	-1.12	0.26
scale(trialn)	-0.08	0.01	-5.18	2.3e-07 (***)
1M	0.28	0.04	7.22	5.1e-13 (***)
2M	0.68	0.05	12.8	1.6e-37 (***)
PC8_L2_History	0.11	0.04	2.38	0.02 (*)
1M*PC8_L2_History	0.04	0.04	0.98	0.33
2M*PC8_L2_History	9.6e-03	0.05	0.18	0.86
Exp. 1 ‘Yes’ responses - PC3 (L2 Proficiency)				
participant response~scale(trialn)+testing_condition*RC3_prof_L2+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.27	0.52		
1M	0.04	0.19	-0.14	
2M	0.24	0.49	-0.38	0.97
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.04	-1.13	0.26
scale(trialn)	-0.08	0.01	-5.17	2.4e-07 (***)
1M	0.28	0.04	7.32	2.5e-13 (***)
2M	0.68	0.05	13.12	2.5e-39 (***)
PC3_L2_Proficiency	-0.07	0.04	-1.56	0.12
1M*PC3_L2_Proficiency	0.08	0.04	2.24	0.03 (*)
2M*PC3_L2_Proficiency	0.15	0.05	2.91	3.6e-03 (**)
Exp. 1 ‘Yes’ responses - Multilingual experience				
participant response~scale(trialn)+testing_condition*scale(multiexp)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.04	-1.23	0.22
scale(trialn)	-0.08	0.01	-5.18	2.2e-07 (***)
1M	0.27	0.04	7.85	4.3e-15 (***)
2M	0.67	0.04	18.44	6.7e-76 (***)
scale(Multilingual experience)	0.03	0.04	0.8	0.42
1M*scale(Multilingual experience)	0.04	0.04	1.27	0.2
2M*scale(Multilingual experience)	0.05	0.04	1.3	0.19

Exp. 1 ‘Yes’ responses - Bilingual balance (L1-L2 difference)				
participant response~testing_condition*scale(L1_L2_diff)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.05	0.04	-1.17	0.24
1M	0.28	0.04	7.86	3.7e-15 (***)
2M	0.67	0.04	18.42	9.5e-76 (***)
scale(L1-L2 difference)	-5.9e-04	0.04	-0.01	0.99
1M*scale(L1-L2 difference)	-0.1	0.04	-2.81	4.9e-03 (**)
2M*scale(L1-L2 difference)	-0.13	0.04	-3.65	2.6e-04 (***)
Exp. 1 ‘Yes’ responses - Multilingual balance (Entropy)				
participant response~testing_condition*ent+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.11	0.14	-0.79	0.43
1M	0.14	0.11	1.31	0.19
2M	0.47	0.11	4.12	3.8e-05 (***)
Entropy	0.04	0.1	0.44	0.66
1M*Entropy	0.1	0.08	1.23	0.22
2M*Entropy	0.15	0.09	1.8	0.07 (.)
Exp. 1 ‘Yes’ responses - Multilingual balance (cosine similarity)				
participant response~testing_condition*cossim+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.13	1.34	-0.1	0.92
1M	0.41	1.1	0.37	0.71
2M	0.26	1.11	0.23	0.82
Cosine similarity	0.09	1.41	0.06	0.95
1M*Cosine similarity	-0.14	1.15	-0.12	0.9
2M*Cosine similarity	0.44	1.16	0.37	0.71
Exp. 1 ‘Yes’ responses - Script				
participant response~scale(trialn)+testing_condition*script+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.04	0.05	-0.91	0.36
scale(trialn)	-0.08	0.01	-5.17	2.3e-07 (***)
1M	0.27	0.04	7.42	1.1e-13 (***)
2M	0.65	0.04	16.8	2.7e-63 (***)
Different script	-0.1	0.14	-0.7	0.49
1M*Different script	-1.2e-03	0.11	-0.01	0.99
2M*Different script	0.25	0.12	2.07	0.04 (*)

Exp. 1 ‘Yes’ responses - Category				
participant response~testing_condition*category+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.11	0.29	0.38	0.7
1M	-0.36	0.24	-1.51	0.13
2M	-0.25	0.24	-1.05	0.3
Bilingual	-0.24	0.3	-0.81	0.42
Trilingual	-0.07	0.3	-0.23	0.82
Quadrilingual	-0.15	0.3	-0.48	0.63
1M*Bilingual	0.71	0.24	2.89	3.8e-03 (**)
2M*Bilingual	1.03	0.25	4.12	3.8e-05 (***)
1M*Trilingual	0.58	0.25	2.34	0.02 (*)
2M*Trilingual	0.81	0.25	3.23	1.2e-03 (**)
1M*Quadrilingual	0.63	0.25	2.55	0.01 (*)
2M*Quadrilingual	0.96	0.25	3.79	1.5e-04 (***)

Exp. 1 2M				
participant response~scale(trialn)+string nature+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.06	11.24	2.6e-29 (***)
scale(trialn)	-0.16	0.03	-5.61	2.1e-08 (***)
within_language	-0.01	0.05	-0.23	0.82
Exp. 1 2M - Gender				
participant response~scale(trialn)+string nature*Gender+(1+string nature sbj_ID)				
Random effects	Variance		SD	Corr
(Intercept)	0.34		0.58	
within_language	0.01		0.1	-1
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.6	0.08	7.85	4.1e-15 (***)
scale(trialn)	-0.16	0.03	-5.6	2.2e-08 (***)
within_language	6.2e-04	0.07	8.5e-03	0.99
Other	0.51	0.48	1.07	0.29
Woman	0.06	0.12	0.53	0.59
within_language*Other	0.58	0.52	1.11	0.27
within_language*Woman	-0.08	0.11	-0.68	0.5
Exp. 1 2M - Age				
participant response~scale(trialn)+string nature*scale(Age)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.06	11.27	1.8e-29 (***)
scale(trialn)	-0.16	0.03	-5.63	1.8e-08 (***)
within_language	-0.01	0.05	-0.23	0.82
scale(Age)	0.03	0.06	0.52	0.6
within_language*scale(Age)	0.04	0.06	0.74	0.46
Exp. 1 2M - PC1 (L3)				
participant response~scale(trialn)+string nature*RC1_L3+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.06	11.24	2.5e-29 (***)
scale(trialn)	-0.16	0.03	-5.6	2.1e-08 (***)
within_language	-0.01	0.06	-0.23	0.82
PC1_L3	0.02	0.06	0.32	0.75
within_language*PC1_L3	3.3e-04	0.06	6.0e-03	1

Exp. 1 2M - PC9 (L4)				
participant response~scale(trialn)+string nature*RC9_L4+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.06	11.24	2.7e-29 (***)
scale(trialn)	-0.16	0.03	-5.61	2.1e-08 (***)
within_language	-0.01	0.05	-0.23	0.82
PC9_L4	-0.04	0.06	-0.77	0.44
within_language*PC9_L4	0.04	0.06	0.77	0.44
Exp. 1 2M - PC2 (L1 vs L2 Use)				
participant response~scale(trialn)+string nature*RC2_use_L1vsL2+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.06	11.27	1.9e-29 (***)
scale(trialn)	-0.16	0.03	-5.61	2.0e-08 (***)
within_language	-0.01	0.05	-0.22	0.82
PC2_L1vsL2_Use	0.07	0.06	1.21	0.23
within_language*PC2_L1vsL2_Use	3.5e-04	0.06	6.3e-03	0.99
Exp. 1 2M - PC8 (L2 History)				
participant response~scale(trialn)+string nature*RC8_hist_L2+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.06	11.32	1.0e-29 (***)
scale(trialn)	-0.16	0.03	-5.62	1.9e-08 (***)
within_language	-0.01	0.06	-0.22	0.82
PC8_L2_History	0.11	0.06	1.91	0.056 (.)
within_language*PC8_L2_History	-1.6e-03	0.06	-0.03	0.98
Exp. 1 2M - PC3 (L2 Proficiency)				
participant response~scale(trialn)+string nature*RC3_prof_L2+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.63	0.06	11.32	1.0e-29 (***)
scale(trialn)	-0.16	0.03	-5.57	2.6e-08 (***)
within_language	-0.01	0.06	-0.27	0.79
PC3_L2_Proficiency	0.13	0.05	2.34	0.02 (*)
within_language*PC3_L2_Proficiency	-0.1	0.05	-1.89	0.058 (.)
Exp. 1 2M - Multilingual experience				
participant response~scale(trialn)+string nature*scale(multiexp)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.06	11.29	1.5e-29 (***)
scale(trialn)	-0.16	0.03	-5.61	2.0e-08 (***)
within_language	-0.01	0.06	-0.22	0.82
scale(Multilingual experience)	0.06	0.06	1.16	0.25
within_language*scale(Multilingual experience)	0.03	0.06	0.47	0.64

Exp. 1 2M - Bilingual Balance (L1-L2 difference)				
participant response~scale(trialn)+string nature*scale(L1_L2_diff)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.05	11.39	4.5e-30 (***)
scale(trialn)	-0.16	0.03	-5.62	1.9e-08 (***)
within_language	-0.01	0.06	-0.22	0.82
scale(L1-L2 difference)	-0.13	0.05	-2.32	0.02 (*)
within_language*scale(L1-L2 difference)	-2.6e-03	0.06	-0.05	0.96
Exp. 1 2M - Multilingual balance (Entropy)				
participant response~scale(trialn)+string nature*ent+(1+string nature subj_ID)				
Random effects	Variance		SD	Corr
(Intercept)	0.33		0.58	
within_language	0.01		0.09	-1
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.42	0.18	2.29	0.02 (*)
scale(trialn)	-0.16	0.03	-5.62	2.0e-08 (***)
within_language	-0.08	0.17	-0.46	0.64
Entropy	0.16	0.14	1.22	0.22
within_language*Entropy	0.04	0.13	0.35	0.73
Exp. 1 2M - Multilingual balance (cosine similarity)				
participant response~scale(trialn)+string nature*scale(cossim)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.06	11.24	2.5e-29 (***)
scale(trialn)	-0.16	0.03	-5.61	2.0e-08 (***)
within_language	-0.01	0.05	-0.22	0.83
scale(Cosine similarity)	3.3e-03	0.06	0.06	0.95
within_language*scale(Cosine similarity)	0.04	0.05	0.64	0.52
Exp. 1 2M - Script				
participant response~scale(trialn)+string nature*script+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.6	0.06	10.31	6.4e-25 (***)
scale(trialn)	-0.16	0.03	-5.59	2.2e-08 (***)
within_language	-2.3e-03	0.06	-0.04	0.97
Different script	0.2	0.18	1.08	0.28
within_language*Different script	-0.1	0.18	-0.54	0.59

Exp. 1 2M - Category participant response~string nature*category+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.27	0.37	-0.74	0.46
within_language	0.23	0.36	0.64	0.52
Bilingual	0.93	0.38	2.46	0.01 (*)
Trilingual	0.91	0.38	2.38	0.02 (*)
Quadrilingual	0.89	0.38	2.31	0.02 (*)
within_language*Bilingual	-0.28	0.37	-0.74	0.46
within_language*Trilingual	-0.29	0.38	-0.76	0.45
within_language*Quadrilingual	-0.16	0.38	-0.41	0.68

Exp. 2 Familiarity - Gender				
correct~scale(trialn)+Gender+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.29	0.04	6.87	6.3e-12 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
Woman	0.03	0.06	0.44	0.66
Exp. 2 Familiarity - Age				
correct~scale(trialn)+scale(Age)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.3	0.03	10.15	3.5e-24 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
scale(Age)	0.04	0.03	1.52	0.13
Exp. 2 Familiarity - PC1 (L4)				
correct~scale(trialn)+RC1_L4+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.3	0.03	10.08	6.5e-24 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
PC1_L4	1.6e-03	0.03	0.05	0.96
Exp. 2 Familiarity - PC3 (L3)				
correct~scale(trialn)+RC3_L3+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.3	0.03	10.13	4.1e-24 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
PC3_L3	-0.04	0.03	-1.3	0.19
Exp. 2 Familiarity - PC2 (L1 vs L2 Use)				
correct~scale(trialn)+RC2_use_L1vsL2+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.3	0.03	10.09	6.2e-24 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
PC2_L1vsL2_Use	0.01	0.03	0.38	0.7
Exp. 2 Familiarity - PC7 (L2 History)				
correct~scale(trialn)+RC7_hist_L2+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.3	0.03	10.09	6.2e-24 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
PC7_L2_History	0.01	0.03	0.38	0.7

Exp. 2 Familiarity - PC9 (L4 Use)				
correct~scale(trialn)+RC9_use_L4+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.3	0.03	10.23	1.5e-24 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
PC4_L4_Use	-0.07	0.03	-2.3	0.02 (*)
Exp. 2 Familiarity - Multilingual experience				
correct~scale(trialn)+scale(multiexp)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.3	0.03	10.12	4.5e-24 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
scale(Multilingual experience)	-0.03	0.03	-1.15	0.25
Exp. 2 Familiarity - Bilingual balance (L1-L2 difference)				
correct~scale(trialn)+scale(L1_L2_diff)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.3	0.03	10.09	6.1e-24 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
scale(L1-L2 difference)	0.01	0.03	0.5	0.62
Exp. 2 Familiarity - Multilingual balance (entropy)				
correct~scale(trialn)+ent+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.41	0.07	5.89	3.9e-09 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
Entropy	-0.09	0.05	-1.71	0.09 (.)
Exp. 2 Familiarity - Multilingual balance (cosine similarity)				
correct~scale(trialn)+cossim+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.59	0.75	-0.79	0.43
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
Cosine similarity	0.93	0.78	1.19	0.23
Exp. 2 Familiarity - Script				
correct~scale(trialn)+script+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.32	0.03	9.84	7.9e-23 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
Different script	-0.11	0.08	-1.33	0.18

Exp. 2 Familiarity - Category				
correct~scale(trialn)+category+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.38	0.09	4.49	7.1e-06 (***)
scale(trialn)	-0.09	0.03	-3.12	1.8e-03 (**)
Bilingual	-0.05	0.1	-0.53	0.59
Trilingual	-0.08	0.1	-0.74	0.46
Quadilingual	-0.19	0.11	-1.83	0.07 (.)

Exp. 2 ‘Yes’ responses - Conditions				
participant response~scale(trialn)+testing_condition+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.37	0.61		
1M	0.1	0.32	-0.61	
2M	0.22	0.46	-0.46	0.95
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.09	0.05	1.75	0.08 (.)
scale(trialn)	-0.08	0.01	-5.56	2.7e-08 (***)
1M	0.25	0.04	5.84	5.1e-09 (***)
2M	0.55	0.05	10.85	2.0e-27 (***)

Exp. 2 ‘Yes’ responses - Gender				
participant response~scale(trialn)+testing_condition*Gender+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.37	0.61		
1M	0.1	0.31	-0.63	
2M	0.21	0.46	-0.47	0.96
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.08	0.07	1.05	0.29
scale(trialn)	-0.08	0.01	-5.52	3.4e-08 (***)
1M	0.17	0.06	2.82	4.9e-03 (**)
2M	0.49	0.07	6.98	2.9e-12 (***)
Woman	0.03	0.1	0.27	0.78
1M*Woman	0.17	0.09	1.99	0.047 (*)
2M*Woman	0.11	0.1	1.14	0.25

Exp. 2 ‘Yes’ responses - Age				
participant response~scale(trialn)+testing_condition*scale(Age)+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.37	0.61		
1M	0.1	0.31	-0.62	
2M	0.2	0.45	-0.47	0.95
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.09	0.05	1.74	0.08 (.)
scale(trialn)	-0.08	0.01	-5.56	2.7e-08 (***)
1M	0.25	0.04	5.89	3.8e-09 (***)
2M	0.55	0.05	10.99	4.1e-28 (***)
scale(Age)	2.6e-03	0.05	0.05	0.96
1M*scale(Age)	0.06	0.04	1.45	0.15
2M*scale(Age)	0.1	0.05	2.02	0.043 (*)

Exp. 2 ‘Yes’ responses - PC1 (L4)				
participant response~scale(trialn)+testing_condition*RC1_L4+(1+testing_condition subj-ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.36	0.6		
1M	0.1	0.32	-0.6	
2M	0.21	0.46	-0.45	0.95
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.09	0.05	1.76	0.08 (.)
scale(trialn)	-0.08	0.01	-5.56	2.7e-08 (***)
1M	0.25	0.04	5.85	5.0e-09 (***)
2M	0.55	0.05	10.87	1.6e-27 (***)
PC1_L4	0.09	0.05	1.79	0.07 (.)
1M*PC1_L4	-0.03	0.04	-0.67	0.5
2M*PC1_L4	-0.05	0.05	-1.01	0.31

Exp. 2 ‘Yes’ responses - PC3 (L3)				
participant response~scale(trialn)+testing_condition*RC3_L3+(1+testing_condition subj-ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.37	0.61		
1M	0.1	0.31	-0.6	
2M	0.22	0.46	-0.46	0.97
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.09	0.05	1.76	0.08 (.)
scale(trialn)	-0.08	0.01	-5.55	2.8e-08 (***)
1M	0.25	0.04	5.87	4.5e-09 (***)
2M	0.55	0.05	10.85	2.0e-27 (***)
PC3_L3	0.06	0.05	1.15	0.25
1M*PC3_L3	-0.07	0.04	-1.74	0.08 (.)
2M*PC3_L3	-0.02	0.05	-0.35	0.72

Exp. 2 ‘Yes’ responses - PC2 (L1 vs L2 Use)				
participant response~scale(trialn)+testing_condition*RC2_use_L1vsL2+(1+testing_ condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.37	0.61		
1M	0.1	0.32	-0.62	
2M	0.21	0.45	-0.48	0.95
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.09	0.05	1.74	0.08 (.)
scale(trialn)	-0.08	0.01	-5.54	3.0e-08 (***)
1M	0.25	0.04	5.86	4.6e-09 (***)
2M	0.55	0.05	10.95	6.5e-28 (***)
PC2_L1vsL2_Use	9.9e-03	0.05	0.19	0.85
1M*PC2_L1vsL2_Use	0.06	0.04	1.33	0.18
2M*PC2_L1vsL2_Use	0.1	0.05	2.03	0.042 (*)
Exp. 2 ‘Yes’ responses - PC7 (L2 History)				
participant response~scale(trialn)+testing_condition*RC7_hist_L2+(1+testing_ condi- tion sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.36	0.6		
1M	0.1	0.32	-0.61	
2M	0.22	0.46	-0.47	0.95
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.09	0.05	1.78	0.07 (.)
scale(trialn)	-0.08	0.01	-5.55	2.8e-08 (***)
1M	0.25	0.04	5.84	5.2e-09 (***)
2M	0.55	0.05	10.84	2.3e-27 (***)
PC7_L2_History	0.11	0.05	2.2	0.03 (*)
1M*PC7_L2_History	-7.4e-03	0.04	-0.17	0.87
2M*PC7_L2_History	-8.3e-03	0.05	-0.16	0.87

Exp. 2 ‘Yes’ responses - PC9 (L4 Use)				
participant response~scale(trialn)+testing_condition*RC9_use.L4+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.37	0.61		
1M	0.1	0.32	-0.61	
2M	0.22	0.46	-0.46	0.96
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.09	0.05	1.75	0.08 (.)
scale(trialn)	-0.08	0.01	-5.56	2.7e-08 (***)
1M	0.25	0.04	5.81	6.3e-09 (***)
2M	0.55	0.05	10.86	1.8e-27 (***)
PC9.L4.Use	-0.01	0.05	-0.24	0.81
1M*PC9.L4.Use	-0.03	0.05	-0.56	0.58
2M*PC9.L4.Use	0.02	0.06	0.34	0.74
Exp. 2 ‘Yes’ responses - Multilingual experience				
participant response~scale(trialn)+testing_condition*scale(multiexp)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.08	0.04	1.85	0.06 (.)
scale(trialn)	-0.08	0.01	-5.44	5.2e-08 (***)
1M	0.26	0.04	7.32	2.5e-13 (***)
2M	0.55	0.04	15.18	5.0e-52 (***)
scale(Multilingual experience)	0.15	0.04	3.34	8.3e-04 (***)
1M*scale(Multilingual experience)	-0.08	0.04	-2.27	0.02 (*)
2M*scale(Multilingual experience)	-0.02	0.04	-0.64	0.52
Exp. 2 ‘Yes’ responses - Bilingual balance (L1-L2 difference)				
participant response~scale(trialn)+testing_condition*scale(L1.L2.diff)+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.36	0.6		
1M	0.1	0.32	-0.6	
2M	0.21	0.46	-0.5	0.98
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.09	0.05	1.74	0.08 (.)
scale(trialn)	-0.08	0.01	-5.53	3.2e-08 (***)
1M	0.25	0.04	5.85	5.0e-09 (***)
2M	0.55	0.05	10.9	1.2e-27 (***)
scale(L1-L2 difference)	-0.1	0.05	-2.03	0.043 (*)
1M*scale(L1-L2 difference)	0.04	0.04	0.92	0.36
2M*scale(L1-L2 difference)	-0.06	0.05	-1.19	0.23

Exp. 2 ‘Yes’ responses - Multilingual balance (Entropy)				
participant response~testing_condition*ent+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.36	0.6		
1M	0.1	0.31	-0.59	
2M	0.21	0.46	-0.48	0.99
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.1	0.12	-0.83	0.41
1M	0.41	0.1	4.18	2.9e-05 (***)
2M	0.51	0.12	4.39	1.1e-05 (***)
Entropy	0.17	0.09	1.8	0.07 (.)
1M*Entropy	-0.14	0.08	-1.85	0.06 (.)
2M*Entropy	0.03	0.09	0.32	0.75
Exp. 2 ‘Yes’ responses - Multilingual balance (cosine similarity)				
participant response~testing_condition*cossim+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-1.38	1.15	-1.2	0.23
1M	-1.39	0.89	-1.56	0.12
2M	-0.07	0.9	-0.08	0.94
Cosine similarity	1.53	1.19	1.28	0.2
1M*Cosine similarity	1.72	0.93	1.85	0.06 (.)
2M*Cosine similarity	0.64	0.94	0.69	0.49
Exp. 2 ‘Yes’ responses - Script				
participant response~testing_condition*script+(1+testing_condition sbj_ID)				
Random effects	Variance	SD	Corr	
(Intercept)	0.37	0.61		
1M	0.1	0.32	-0.6	
2M	0.21	0.46	-0.47	0.99
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.08	0.06	1.44	0.15
1M	0.28	0.05	6	2.0e-09 (***)
2M	0.52	0.05	9.63	6.2e-22 (***)
Different script	0.09	0.14	0.59	0.55
1M*Different script	-0.2	0.12	-1.67	0.09 (.)
2M*Different script	0.13	0.14	0.9	0.37

Exp. 2 ‘Yes’ responses - Category				
participant response~testing_condition*category+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.04	0.13	0.31	0.76
1M	0.4	0.1	3.9	9.8e-05 (***)
2M	0.41	0.1	4.01	6.0e-05 (***)
Bilingual	-0.05	0.15	-0.31	0.76
Trilingual	0.08	0.16	0.51	0.61
Quadrilingual	0.2	0.16	1.2	0.23
1M*Bilingual	-0.08	0.12	-0.66	0.51
2M*Bilingual	0.21	0.12	1.77	0.08 (.)
1M*Trilingual	-0.21	0.12	-1.68	0.09 (.)
2M*Trilingual	0.13	0.12	1.03	0.3
1M*Quadrilingual	-0.25	0.13	-1.93	0.054 (.)
2M*Quadrilingual	0.09	0.13	0.7	0.49

Exp. 2 2M				
participant response~scale(trialn)+string nature+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.36	0.6	
	within_language	0.01	0.09	-0.64
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.66	0.06	11.19	4.8e-29 (***)
scale(trialn)	-0.11	0.03	-4.24	2.2e-05 (***)
within_language	-0.06	0.05	-1.07	0.29

Exp. 2 2M - Gender				
participant response~scale(trialn)+string nature*Gender+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.36	0.6	
	within_language	0.01	0.09	-0.66
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.6	0.08	7.32	2.5e-13 (***)
scale(trialn)	-0.11	0.03	-4.24	2.3e-05 (***)
within_language	-0.07	0.07	-0.95	0.34
Woman	0.12	0.12	1.02	0.31
within_language*Woman	0.03	0.11	0.25	0.81

Exp. 2 2M - Age				
participant response~scale(trialn)+string nature*scale(Age)+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.35	0.59	
	within_language	0.01	0.09	-0.67
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.66	0.06	11.26	2.2e-29 (***)
scale(trialn)	-0.11	0.03	-4.24	2.2e-05 (***)
within_language	-0.06	0.05	-1.07	0.28
scale(Age)	0.1	0.06	1.68	0.09 (.)
within_language*scale(Age)	4.9e-03	0.05	0.09	0.93

Exp. 2 2M - PC1 (L4)				
participant response~scale(trialn)+string nature*RC1_L4+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.36	0.6	
	within_language	0.01	0.09	-0.65
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.66	0.06	11.2	4.0e-29 (***)
scale(trialn)	-0.11	0.03	-4.24	2.2e-05 (***)
within_language	-0.06	0.05	-1.07	0.29
PC1_L4	0.04	0.06	0.69	0.49
within_language*PC1_L4	1.4e-03	0.05	0.03	0.98

Exp. 2 2M - PC3 (L3)				
participant response~scale(trialn)+string nature*RC3_L3+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.36	0.6	
	within_language	0.01	0.08	-0.73
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.66	0.06	11.19	4.6e-29 (***)
scale(trialn)	-0.11	0.03	-4.25	2.2e-05 (***)
within_language	-0.06	0.05	-1.06	0.29
PC3_L3	0.02	0.06	0.42	0.67
within_language*PC3_L3	0.03	0.05	0.62	0.54
Exp. 2 2M - PC2 (L1 vs L2 Use)				
participant response~scale(trialn)+string nature*RC2_use_L1vsL2+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.34	0.58	
	within_language	0.01	0.09	-0.63
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.66	0.06	11.35	7.7e-30 (***)
scale(trialn)	-0.11	0.03	-4.23	2.4e-05 (***)
within_language	-0.06	0.05	-1.06	0.29
PC2_L1vsL2_Use	0.13	0.06	2.25	0.02 (*)
within_language*PC2_L1vsL2_Use	-0.02	0.05	-0.35	0.72
Exp. 2 2M - PC7 (L2 History)				
participant response~scale(trialn)+string nature*RC7_hist_L2+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.35	0.59	
	within_language	0.01	0.09	-0.61
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.66	0.06	11.33	9.7e-30 (***)
scale(trialn)	-0.11	0.03	-4.24	2.2e-05 (***)
within_language	-0.06	0.05	-1.08	0.28
PC7_L2_History	0.11	0.06	1.87	0.06 (.)
within_language*PC7_L2_History	-0.02	0.05	-0.38	0.71

Exp. 2 2M - PC9 (L4 Use)				
participant response~scale(trialn)+string nature*RC9_use_L4+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.36	0.6	
	within_language	0.01	0.09	-0.64
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.66	0.06	11.19	4.7e-29 (***)
scale(trialn)	-0.11	0.03	-4.24	2.3e-05 (***)
within_language	-0.06	0.05	-1.07	0.28
PC9_L4_Use	0.01	0.06	0.18	0.86
within_language*PC9_L4_Use	-7.9e-03	0.06	-0.13	0.9
Exp. 2 2M - Multilingual experience				
participant response~scale(trialn)+string nature*scale(multiexp)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.65	0.06	11.62	3.1e-31 (***)
scale(trialn)	-0.11	0.03	-4.24	2.2e-05 (***)
within_language	-0.05	0.05	-0.9	0.37
scale(Multilingual experience)	0.09	0.06	1.7	0.09 (.)
within_language*scale(Multilingual experience)	0.06	0.05	1.14	0.25
Exp. 2 2M - Bilingual Balance (L1-L2 difference)				
participant response~scale(trialn)+string nature*scale(L1_L2_diff)+(1+string nature sbj_ID)				
Random effects		Variance	SD	Corr
	(Intercept)	0.34	0.58	
	within_language	0.01	0.08	-0.76
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.66	0.06	11.39	4.7e-30 (***)
scale(trialn)	-0.11	0.03	-4.19	2.8e-05 (***)
within_language	-0.06	0.05	-1.06	0.29
scale(L1-L2 difference)	-0.15	0.06	-2.69	7.1e-03 (**)
within_language*scale(L1-L2 difference)	-0.02	0.05	-0.39	0.7

Exp. 2 2M - Multilingual balance (Entropy)				
participant response~scale(trialn)+string nature*ent+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.52	0.13	3.99	6.7e-05 (***)
scale(trialn)	-0.11	0.03	-4.25	2.2e-05 (***)
within_language	-0.22	0.12	-1.87	0.06 (.)
Entropy	0.12	0.1	1.15	0.25
within_language*Entropy	0.15	0.09	1.62	0.1

Exp. 2 2M - Multilingual balance (cosine similarity)				
participant response~scale(trialn)+string nature*scale(cossim)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.65	0.06	11.55	7.3e-31 (***)
scale(trialn)	-0.11	0.03	-4.23	2.3e-05 (***)
within_language	-0.05	0.05	-0.98	0.33
scale(Cosine similarity)	0.14	0.06	2.45	0.01 (*)
within_language*scale(Cosine similarity)	-0.11	0.05	-2.13	0.03 (*)

Exp. 2 2M - Script				
participant response~scale(trialn)+string nature*script+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.64	0.06	10.47	1.1e-25 (***)
scale(trialn)	-0.11	0.03	-4.23	2.3e-05 (***)
within_language	-0.1	0.06	-1.68	0.09 (.)
Different script	0.06	0.16	0.39	0.7
within_language*Different script	0.32	0.15	2.12	0.03 (*)

Exp. 2 2M - Category				
participant response~scale(trialn)+string nature*category+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.61	0.16	3.76	1.7e-04 (***)
scale(trialn)	-0.11	0.03	-4.22	2.4e-05 (***)
within_language	-0.33	0.15	-2.23	0.03 (*)
Bilingual	0.01	0.19	0.07	0.94
Trilingual	0.09	0.19	0.47	0.64
Quadrilingual	0.07	0.2	0.33	0.74
within_language*Bilingual	0.32	0.17	1.85	0.06 (.)
within_language*Trilingual	0.23	0.18	1.31	0.19
within_language*Quadrilingual	0.45	0.19	2.38	0.02 (*)

Exp. 3 Familiarity - Gender				
correct~scale(trialn)+Gender+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.64	0.05	13.7	1.0e-42 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
Woman	0.09	0.09	1	0.32
Exp. 3 Familiarity - Age				
correct~scale(trialn)+scale(Age)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.67	0.04	16.85	1.1e-63 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
scale(Age)	0.03	0.04	0.81	0.42
Exp. 3 Familiarity - PC1 (L4)				
correct~scale(trialn)+RC1_L4+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.67	0.04	16.86	9.6e-64 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
PC1_L4	0.04	0.04	0.9	0.37
Exp. 3 Familiarity - PC3 (L3)				
correct~scale(trialn)+RC3_L3+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.67	0.04	16.82	1.7e-63 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
PC3_L3	1.7e-03	0.04	0.04	0.96
Exp. 3 Familiarity - PC2 (L1 vs L2 Use)				
correct~scale(trialn)+RC2_use_L1vsL2+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.67	0.04	16.85	1.0e-63 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
PC2_L1vsL2_Use	0.03	0.04	0.86	0.39
Exp. 3 Familiarity - PC8 (L2 History)				
correct~scale(trialn)+RC8_hist_L2+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.67	0.04	16.87	7.0e-64 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
PC8_L2_History	-0.04	0.04	-1.05	0.29

Exp. 3 Familiarity - Multilingual experience				
correct~scale(trialn)+scale(multiexp)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.67	0.04	16.85	1.1e-63 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
scale(Multilingual experience)	0.03	0.04	0.73	0.47
Exp. 3 Familiarity - Bilingual balance (L1-L2 difference)				
correct~scale(trialn)+scale(L1_L2_diff)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.67	0.04	16.83	1.4e-63 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
scale(L1-L2 difference)	-0.02	0.04	-0.52	0.61
Exp. 3 Familiarity - Multilingual balance (entropy)				
correct~scale(trialn)+ent+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.59	0.09	6.45	1.1e-10 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
Entropy	0.07	0.08	0.87	0.38
Exp. 3 Familiarity - Multilingual balance (cosine similarity)				
correct~scale(trialn)+cossim+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	1.45	1.07	1.36	0.17
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
Cosine similarity	-0.81	1.11	-0.73	0.46
Exp. 3 Familiarity - Script				
correct~scale(trialn)+script+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.63	0.06	10.85	1.9e-27 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
Different script	0.07	0.08	0.9	0.37
Exp. 3 Familiarity - Category				
correct~scale(trialn)+category+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.63	0.11	5.51	3.6e-08 (***)
scale(trialn)	-0.1	0.03	-3.68	2.4e-04 (***)
Bilingual	1.4e-03	0.13	0.01	0.99
Trilingual	0.09	0.14	0.69	0.49
Quadrilingual	0.06	0.16	0.39	0.7

Exp. 3 ‘Yes’ responses - Conditions				
participant response~scale(trialn)+testing_condition+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.5	0.05	-10.96	6.2e-28 (***)
scale(trialn)	-0.06	0.01	-4.01	6.0e-05 (***)
1M	0.63	0.03	18.16	1.0e-73 (***)
2M	1.25	0.04	35.22	1.0e-271 (***)
Exp. 3 ‘Yes’ responses - Gender				
participant response~scale(trialn)+testing_condition*Gender+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.43	0.05	-7.99	1.3e-15 (***)
scale(trialn)	-0.06	0.01	-3.98	7.0e-05 (***)
1M	0.58	0.04	14.3	2.1e-46 (***)
2M	1.2	0.04	28.52	7.0e-179 (***)
Woman	-0.25	0.1	-2.45	0.01 (*)
1M*Woman	0.15	0.08	1.98	0.047 (*)
2M*Woman	0.18	0.08	2.29	0.02 (*)
Exp. 3 ‘Yes’ responses - Age				
participant response~scale(trialn)+testing_condition*scale(Age)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.5	0.05	-10.97	5.2e-28 (***)
scale(trialn)	-0.06	0.01	-4.03	5.6e-05 (***)
1M	0.63	0.03	18.17	9.5e-74 (***)
2M	1.25	0.04	35.24	5.7e-272 (***)
scale(Age)	9.6e-03	0.05	0.21	0.83
1M*scale(Age)	0.03	0.03	0.79	0.43
2M*scale(Age)	0.06	0.04	1.68	0.09 (.)

Exp. 3 ‘Yes’ responses - PC1 (L4)				
participant response~scale(trialn)+testing_condition*RC1_L4+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.5	0.05	-10.93	7.9e-28 (***)
scale(trialn)	-0.06	0.01	-4.03	5.5e-05 (***)
1M	0.63	0.03	18.15	1.3e-73 (***)
2M	1.25	0.04	35.2	1.7e-271 (***)
PC1_L4	0.05	0.05	1.08	0.28
1M*PC1_L4	-0.07	0.04	-1.96	0.05 (*)
2M*PC1_L4	-0.11	0.04	-3.05	2.3e-03 (**)

Exp. 3 ‘Yes’ responses - PC3 (L3)				
participant response~scale(trialn)+testing_condition*RC3_L3+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.5	0.05	-11.04	2.6e-28 (***)
scale(trialn)	-0.06	0.01	-3.99	6.7e-05 (***)
1M	0.63	0.03	18.27	1.4e-74 (***)
2M	1.26	0.04	35.31	4.7e-273 (***)
PC3_L3	-0.03	0.05	-0.73	0.47
1M*PC3_L3	0.12	0.04	3.32	9.1e-04 (***)
2M*PC3_L3	0.13	0.04	3.52	4.3e-04 (***)

Exp. 3 ‘Yes’ responses - PC2 (L1 vs L2 Use)				
participant response~scale(trialn)+testing_condition*RC2_use_L1vsL2+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.5	0.05	-10.99	4.2e-28 (***)
scale(trialn)	-0.06	0.01	-3.88	1.0e-04 (***)
1M	0.62	0.03	18.13	1.8e-73 (***)
2M	1.25	0.04	35.2	2.0e-271 (***)
PC2_L1vsL2_Use	-0.03	0.05	-0.73	0.47
1M*PC2_L1vsL2_Use	0.1	0.03	2.89	3.8e-03 (**)
2M*PC2_L1vsL2_Use	0.19	0.04	5.25	1.5e-07 (***)

Exp. 3 ‘Yes’ responses - PC8 (L2 History)				
participant response~scale(trialn)+testing_condition*RC8_hist_L2+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.5	0.05	-11.06	2.0e-28 (***)
scale(trialn)	-0.06	0.01	-4.01	6.2e-05 (***)
1M	0.63	0.03	18.16	1.1e-73 (***)
2M	1.25	0.04	35.21	1.3e-271 (***)
PC8_L2_History	0.07	0.04	1.45	0.15
1M*PC8_L2_History	0.04	0.03	1.22	0.22
2M*PC8_L2_History	0.02	0.04	0.43	0.67

Exp. 3 ‘Yes’ responses - Multilingual experience				
participant response~scale(trialn)+testing_condition*scale(multiexp)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.5	0.05	-11.07	1.7e-28 (***)
scale(trialn)	-0.06	0.01	-3.98	6.8e-05 (***)
1M	0.62	0.03	18.14	1.5e-73 (***)
2M	1.25	0.04	35.22	1.1e-271 (***)
scale(Multilingual experience)	0.03	0.04	0.67	0.5
1M*scale(Multilingual experience)	0.06	0.03	1.88	0.06 (.)
2M*scale(Multilingual experience)	0.1	0.04	2.71	6.7e-03 (**)

Exp. 3 ‘Yes’ responses - Bilingual balance (L1-L2 difference)				
participant response~scale(trialn)+testing_condition*scale(L1_L2_diff)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.5	0.05	-11.05	2.1e-28 (***)
scale(trialn)	-0.06	0.01	-3.95	7.9e-05 (***)
1M	0.62	0.03	18.14	1.6e-73 (***)
2M	1.25	0.04	35.22	9.5e-272 (***)
scale(L1-L2 difference)	-3.5e-03	0.04	-0.08	0.94
1M*scale(L1-L2 difference)	-0.1	0.03	-2.77	5.6e-03 (**)
2M*scale(L1-L2 difference)	-0.15	0.04	-4.2	2.7e-05 (***)

Exp. 3 ‘Yes’ responses - Multilingual balance (Entropy)				
participant response~scale(trialn)+testing_condition*ent+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.41	0.11	-3.86	1.1e-04 (***)
scale(trialn)	-0.06	0.01	-3.96	7.4e-05 (***)
1M	0.43	0.08	5.32	1.1e-07 (***)
2M	0.95	0.08	11.55	7.5e-31 (***)
Entropy	-0.08	0.09	-0.93	0.35
1M*Entropy	0.18	0.07	2.71	6.8e-03 (**)
2M*Entropy	0.28	0.07	4.04	5.4e-05 (***)
Exp. 3 ‘Yes’ responses - Multilingual balance (cosine similarity)				
participant response~testing_condition*cossim+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-3.67	1.22	-3.02	2.5e-03 (**)
1M	1.4	0.93	1.5	0.13
2M	2.87	0.95	3.01	2.6e-03 (**)
Cosine similarity	3.3	1.26	2.61	9.0e-03 (**)
1M*Cosine similarity	-0.8	0.97	-0.83	0.41
2M*Cosine similarity	-1.69	0.99	-1.7	0.09 (.)
Exp. 3 ‘Yes’ responses - Script				
participant response~scale(trialn)+testing_condition*script+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.57	0.07	-8.54	1.4e-17 (***)
scale(trialn)	-0.06	0.01	-4	6.4e-05 (***)
1M	0.57	0.05	11.21	3.6e-29 (***)
2M	1.19	0.05	22.81	3.6e-115 (***)
Different script	0.13	0.09	1.42	0.16
1M*Different script	0.1	0.07	1.44	0.15
2M*Different script	0.12	0.07	1.62	0.11
Exp. 3 ‘Yes’ responses - Category				
participant response~testing_condition*category+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	-0.38	0.13	-2.89	3.9e-03 (**)
1M	0.44	0.1	4.37	1.3e-05 (***)
2M	0.95	0.1	9.3	1.4e-20 (***)
Bilingual	-0.1	0.15	-0.7	0.48
Trilingual	-0.15	0.16	-0.94	0.35
Quadrilingual	-0.22	0.18	-1.22	0.22
1M*Bilingual	0.13	0.11	1.18	0.24
2M*Bilingual	0.24	0.11	2.07	0.04 (*)
1M*Trilingual	0.36	0.12	3.01	2.6e-03 (**)
2M*Trilingual	0.54	0.12	4.39	1.1e-05 (***)
1M*Quadrilingual	0.19	0.14	1.34	0.18
2M*Quadrilingual	0.32	0.14	2.25	0.02 (*)

Exp. 3 2M				
participant response~scale(trialn)+string nature+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	11.98	4.4e-33 (***)
scale(trialn)	-0.12	0.03	-4.7	2.6e-06 (***)
within_language	0.13	0.05	2.45	0.01 (*)
Exp. 3 2M - Gender				
participant response~scale(trialn)+string nature*Gender+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.72	0.07	10.56	4.4e-26 (***)
scale(trialn)	-0.12	0.03	-4.69	2.7e-06 (***)
within_language	0.09	0.06	1.49	0.13
Woman	-0.11	0.13	-0.88	0.38
within_language*Woman	0.12	0.11	1.05	0.29
Exp. 3 2M - Age				
participant response~scale(trialn)+string nature*scale(Age)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	12.01	3.1e-33 (***)
scale(trialn)	-0.12	0.03	-4.72	2.4e-06 (***)
within_language	0.13	0.05	2.49	0.01 (*)
scale(Age)	0.04	0.06	0.76	0.45
within_language*scale(Age)	0.06	0.05	1.23	0.22
Exp. 3 2M - PC1 (L4)				
participant response~scale(trialn)+string nature*RC1_L4+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	12	3.4e-33 (***)
scale(trialn)	-0.12	0.03	-4.68	2.8e-06 (***)
within_language	0.13	0.05	2.45	0.01 (*)
PC1_L4	-0.04	0.06	-0.68	0.5
within_language*PC1_L4	-0.03	0.05	-0.5	0.61

Exp. 3 2M - PC3 (L3)				
participant response~scale(trialn)+string nature*RC3_L3+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.7	0.06	12.1	1.0e-33 (***)
scale(trialn)	-0.12	0.03	-4.67	3.0e-06 (***)
within_language	0.12	0.05	2.36	0.02 (*)
PC3_L3	0.11	0.06	1.95	0.051 (.)
within_language*PC3_L3	-0.07	0.05	-1.23	0.22

Exp. 3 2M - PC2 (L1 vs L2 Use)				
participant response~scale(trialn)+string nature*RC2_use_L1vsL2+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	12.23	2.2e-34 (***)
scale(trialn)	-0.12	0.03	-4.65	3.2e-06 (***)
within_language	0.12	0.05	2.43	0.02 (*)
PC2_L1vsL2_Use	0.18	0.06	3.1	2.0e-03 (**)
within_language*PC2_L1vsL2_Use	-0.02	0.05	-0.42	0.67

Exp. 3 2M - PC8 (L2 History)				
participant response~scale(trialn)+string nature*RC8_hist_L2+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	12.05	2.0e-33 (***)
scale(trialn)	-0.12	0.03	-4.71	2.5e-06 (***)
within_language	0.13	0.05	2.44	0.01 (*)
PC8_L2_History	0.09	0.06	1.64	0.1
within_language*PC8_L2_History	-0.03	0.05	-0.49	0.62

Exp. 3 2M - Multilingual experience				
participant response~scale(trialn)+string nature*scale(multiexp)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	12.14	6.7e-34 (***)
scale(trialn)	-0.12	0.03	-4.68	2.9e-06 (***)
within_language	0.12	0.05	2.42	0.02 (*)
scale(Multilingual experience)	0.15	0.06	2.58	9.8e-03 (**)
within_language*scale(Multilingual experience)	-0.04	0.05	-0.69	0.49

Exp. 3 2M - Bilingual Balance (L1-L2 difference)				
participant response~scale(trialn)+string nature*scale(L1_L2_diff)+(1 sbj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	12.22	2.3e-34 (***)
scale(trialn)	-0.12	0.03	-4.68	2.8e-06 (***)
within_language	0.13	0.05	2.45	0.01 (*)
scale(L1-L2 difference)	-0.15	0.06	-2.74	6.2e-03 (**)
within_language*scale(L1-L2 difference)	-5.3e-03	0.05	-0.1	0.92

Exp. 3 2M - Multilingual balance (Entropy)				
participant response~scale(trialn)+string nature*ent+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.48	0.13	3.65	2.6e-04 (***)
scale(trialn)	-0.12	0.03	-4.69	2.7e-06 (***)
within_language	0.1	0.12	0.86	0.39
Entropy	0.19	0.11	1.72	0.09 (.)
within_language*Entropy	0.02	0.1	0.22	0.82
Exp. 3 2M - Multilingual balance (cosine similarity)				
participant response~scale(trialn)+string nature*scale(cossim)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	12.02	2.6e-33 (***)
scale(trialn)	-0.12	0.03	-4.72	2.3e-06 (***)
within_language	0.12	0.05	2.39	0.02 (*)
scale(Cosine similarity)	0.11	0.06	1.99	0.047 (*)
within_language*scale(Cosine similarity)	-0.13	0.05	-2.56	0.01 (*)
Exp. 3 2M - Multilingual balance (cosine similarity - without monolinguals)				
participant response~scale(trialn)+string nature*scale(cossim)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.7	0.06	11.76	6.6e-32 (***)
scale(trialn)	-0.13	0.03	-4.82	1.4e-06 (***)
within_language	0.16	0.05	2.84	4.5e-03 (**)
scale(Cosine similarity)	0.14	0.06	2.42	0.02 (*)
within_language*scale(Cosine similarity)	-0.11	0.05	-2.02	0.044 (*)

Exp. 3 2M - Multilingual balance (use cosine similarity)				
participant response~scale(trialn)+string nature*scale(use_cossim)+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.69	0.06	12.02	2.6e-33 (***)
scale(trialn)	-0.12	0.03	-4.71	2.5e-06 (***)
within_language	0.12	0.05	2.4	0.02 (*)
scale(Use cosine similarity)	0.11	0.06	1.98	0.048 (*)
within_language*scale(Use cosine similarity)	-0.11	0.05	-2.24	0.03 (*)

Exp. 3 2M - Script				
participant response~scale(trialn)+string nature*script+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.56	0.08	6.71	2.0e-11 (***)
scale(trialn)	-0.12	0.03	-4.68	2.8e-06 (***)
within_language	0.13	0.07	1.68	0.09 (.)
Different script	0.24	0.11	2.14	0.03 (*)
within_language*Different script	1.3e-03	0.1	0.01	0.99

Exp. 3 2M - Category				
participant response~scale(trialn)+string nature*category+(1 subj_ID)				
Fixed effects	β	SE	z-value	p-value
(Intercept)	0.62	0.16	3.76	1.7e-04 (***)
scale(trialn)	-0.12	0.03	-4.69	2.7e-06 (***)
within_language	-0.12	0.15	-0.82	0.41
Bilingual	-0.02	0.18	-0.12	0.9
Trilingual	0.28	0.2	1.43	0.15
Quadrilingual	0.04	0.23	0.17	0.86
within_language*Bilingual	0.34	0.16	2.08	0.04 (*)
within_language*Trilingual	0.22	0.18	1.24	0.21
within_language*Quadrilingual	0.16	0.21	0.79	0.43

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